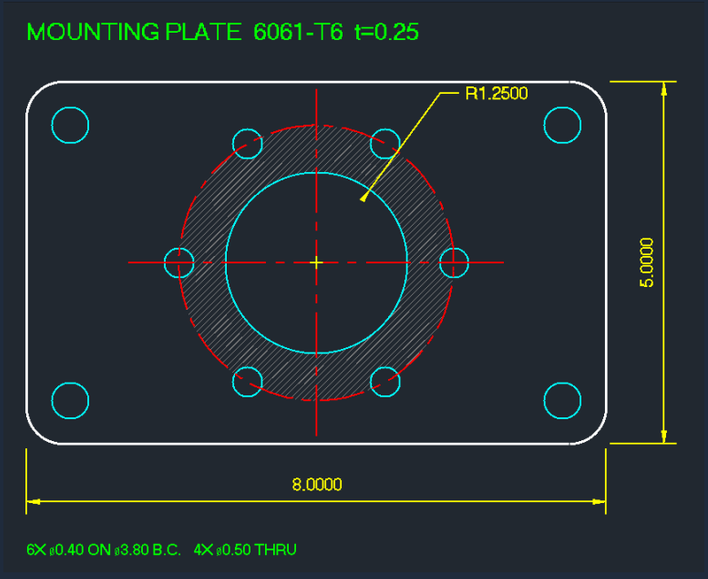
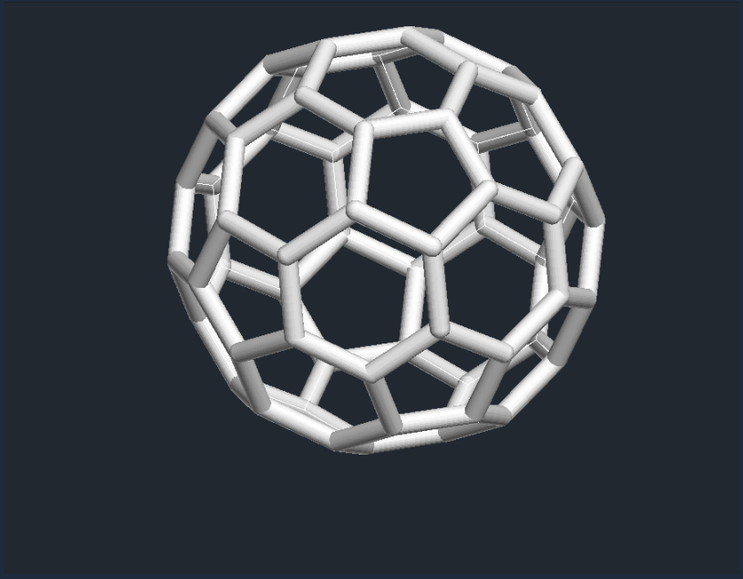
# Fancy Pencil

## User Manual

Drafting with AutoCAD-style commands and AutoLISP, in 2D and 3D, on Linux.

**Version 0.4.24**



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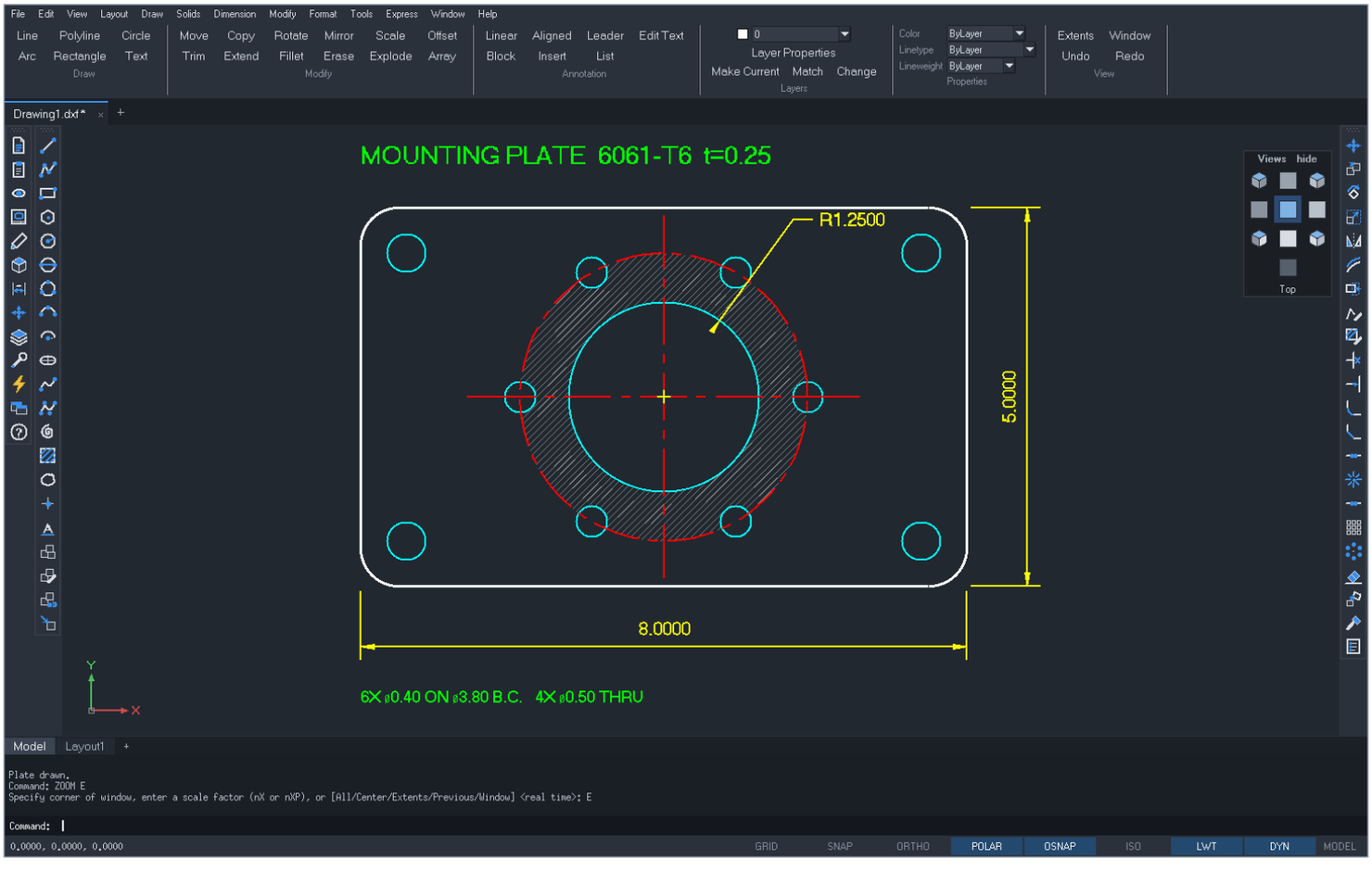
CHAPTER 1

# Welcome to Fancy Pencil

*A drafting program that works like AutoCAD: the same commands, the same prompts, and your AutoLISP.*

Fancy Pencil is a 2D and 3D drafting program for Linux. If you have used AutoCAD, you already know how to use it: you type `LINE` or `L`, you answer the same prompts, you pick points with object snaps, you select objects first or after, and you save drawings that AutoCAD opens. Layers, blocks, dimensions, hatches, text styles, layouts and plotting all work the way you expect.

It also runs **AutoLISP**. Routines written for AutoCAD - `defun c:...`, `command`, `entget`, `ssget` filters, `getkword` and `*error*` handlers - run unchanged in most cases, and you can type AutoLISP straight at the command line.




*Fancy Pencil with a drawing made by an AutoLISP program (the demo plate).*

## What it can do

- **Draw** lines, polylines, rectangles, polygons, circles, arcs, ellipses, splines, helixes, points, revision clouds and hatches.
- **Edit** with Move, Copy, Rotate, Scale, Mirror, Offset, Trim, Extend, Fillet, Chamfer, Array, Stretch, Align, Explode and grips.
- **Annotate** with single-line and paragraph text, dimensions, leaders and multileaders, in text and dimension styles.
- **Organise** with layers, linetypes, lineweights, colors, blocks and draw order.
- **Model in 3D**: boxes, cylinders, spheres and more; extrude, revolve, sweep, loft, press-pull; union, subtract, intersect and slice; the Platonic solids and wireframe struts.
- **Look around** in Top, Front, Side and isometric views, orbit, shade, and split the screen into viewports.
- **Print** to PDF or any printer from the model or from layouts with viewports, at true scale.
- **Exchange** drawings with AutoCAD as DXF, and as DWG with the free ODA File Converter.

## How to read this manual

Chapters 2 to 4 get you started: installing, the window, and how commands are given. The chapters after them each cover one kind of work. Chapter 19 lists the keyboard shortcuts, and the Command Reference at the end lists every command with its aliases.

This manual is also inside Fancy Pencil: press `F1` or choose **Help > User Manual** to open it on screen.

Things you type are shown like `this`. A command's alias (its short name) is shown in brackets: LINE (`L`). Menu paths are written with arrows: **Format > Layer**. Keys are written `Ctrl+S`, `F3`.

CHAPTER 2

# Installing Fancy Pencil

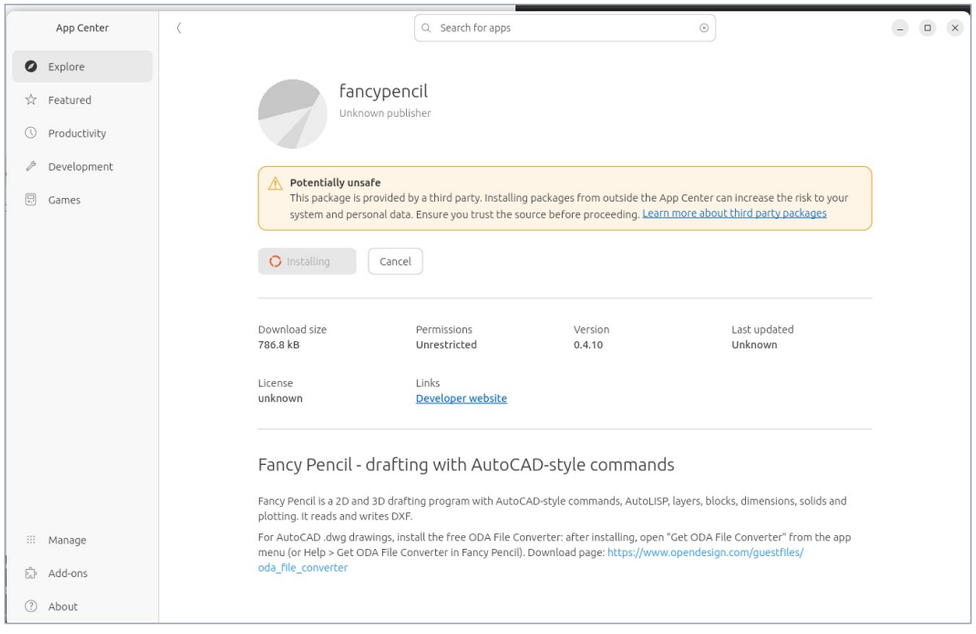
*One file installs everything: Install Fancy Pencil.deb, from the Fancy Pencil Kit.*

## What you need

- A 64-bit computer with **Ubuntu** 22.04 or newer, **Linux Mint** 21 or newer, or **Debian** 12.
- The **Fancy Pencil Kit** folder (on a USB stick or drive, or downloaded). It holds `Install Fancy Pencil.deb` and `INSTRUCTIONS.txt`.
- Your password, to install software.

## Install

1. Open the Fancy Pencil Kit folder and double-click **Install Fancy Pencil.deb**.
2. If it opens in Archive Manager instead (a list of folders called DEBIAN, opt and usr), close that window, right-click the file and choose **Open With... > App Center**.
3. App Center shows Fancy Pencil. Click **Install** and enter your password. App Center also installs SBCL and Tk, the two programs Fancy Pencil runs on.
4. When it has finished, press the Super (Windows) key, type **Fancy Pencil**, and click its icon. Right-click the icon in the dock and choose **Pin to Dash** to keep it there.




*App Center installing Fancy Pencil from the Kit.*

> **Note** App Center shows **Potentially unsafe** and **Unknown publisher** for every program that comes from a file instead of its own store. It is not a fault in Fancy Pencil. Install it if you trust where the Kit came from.

## If App Center won't open the file

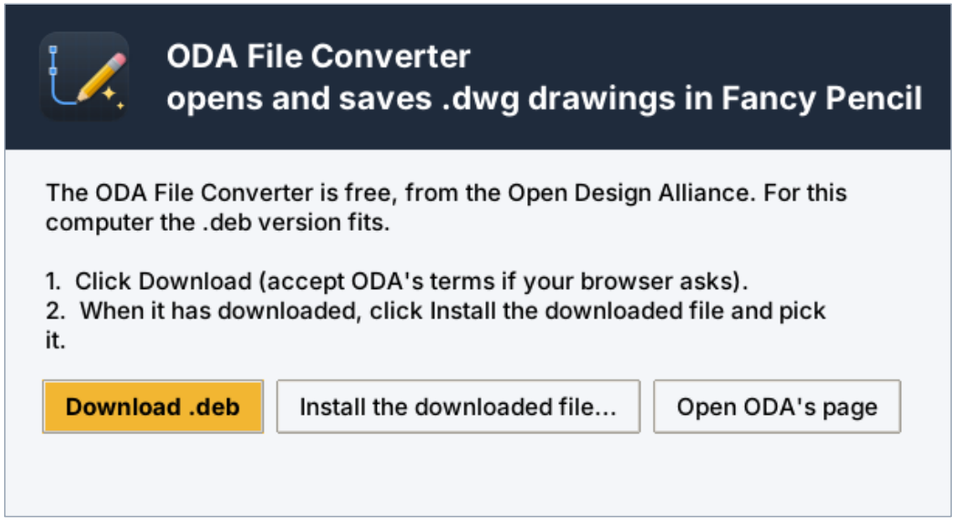
Right-click an empty part of the Kit folder, choose **Open in Terminal**, and type:

```
sudo apt install "./Install Fancy Pencil.deb"
```

## AutoCAD DWG files: ODA File Converter

Fancy Pencil reads and writes AutoCAD's DXF files by itself. For **.dwg** files it uses the free **ODA File Converter** from the Open Design Alliance. To install it:

1. Open **Get ODA File Converter** from the app menu (or, in Fancy Pencil, **Help > Get ODA File Converter**).
2. Click **Download .deb**. Your web browser downloads the converter; accept ODA's terms if it asks.
3. When the download has finished, click **Install the downloaded file...**, pick the file whose name starts with `ODAFileConverter` and ends in `.deb`, and enter your password.




*The Get ODA File Converter window.*

Download page: `www.opendesign.com/guestfiles/oda_file_converter`. Fancy Pencil finds the converter by itself once it is installed.

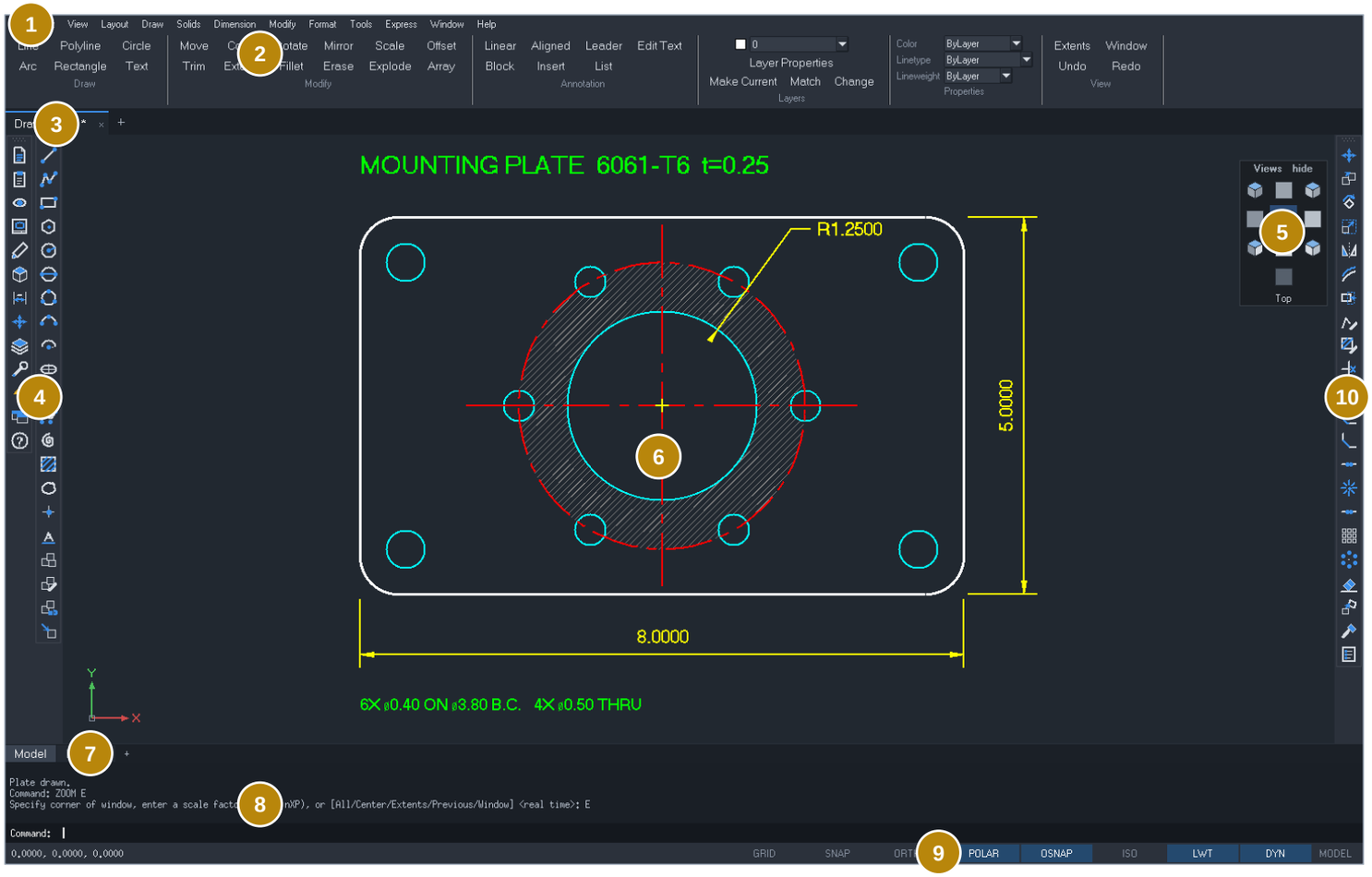
## Updating and removing

- **Update**: double-click a newer Install Fancy Pencil.deb and click **Install** (or **Update**). Your settings, toolbars and drawings are kept.
- **Remove**: in App Center, **Manage**, find fancypencil and remove it; or type `sudo apt remove fancypencil`.
- **Your settings** live in the hidden folder `~/.config/fancypencil`: preferences, recent files, aliases, toolbars and plot settings. Copy it to another computer to bring them along.

CHAPTER 3

# The Fancy Pencil Window

*Where everything is.*




*The parts of the window.*

| | | |
|---|---|---|
| 1 | **Menu bar** | Every command, by subject: File, Edit, View, Layout, Draw, Solids, Dimension, Modify, Format, Tools, Express, Window, Help. |
| 2 | **Ribbon** | The everyday commands in groups (Draw, Modify, Annotation, Layers, Properties, View), the current layer list and the color, linetype and lineweight lists. |
| 3 | **Drawing tabs** | One tab for each open drawing. Click to switch, x to close, + for a new drawing. |
| 4 | **Toolbars** | Icons for commands. The Menus toolbar opens any menu's whole tree; Draw is beside it. Rest the mouse on an icon to see its name. |
| 5 | **View picker** | Click a cube to see the drawing from the Top, a side, or an isometric corner. |

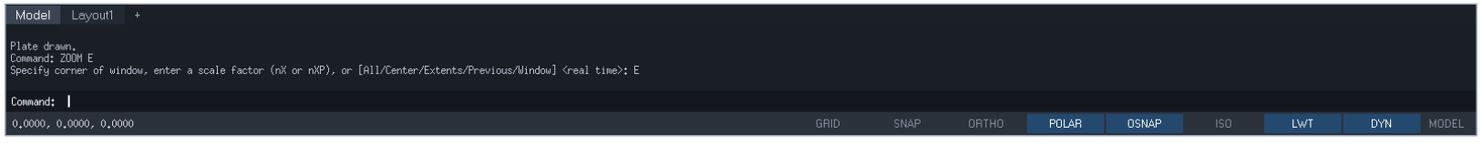
| | | |
|---|---|---|
| 6 | **Drawing area** | Where you draw. Wheel zooms at the cursor, middle-drag pans, middle double-click shows everything. |
| 7 | **Model and layout tabs** | Model is where you draw full size; layouts are sheets of paper for plotting. |
| 8 | **Command line** | Where commands are typed and prompts appear. The history above it shows what happened. |
| 9 | **Status bar** | The cursor's coordinates, and buttons for GRID, SNAP, ORTHO, POLAR, OSNAP, ISO, LWT and DYN. |
| 10 | **Modify toolbar** | Move, Copy, Rotate, Scale, Mirror, Trim, Extend, Fillet, Array and the other editing commands. |

## The ribbon




*The ribbon: click a word to run that command.*

## Command line and status bar




*Prompts appear here; the status bar buttons turn drawing aids on and off.*

## The status bar buttons

| | | |
|---|---|---|
| **GRID** | `F7` | A grid of dots or lines to help you judge sizes. It doesn't print. |
| **SNAP** | `F9` | The cursor jumps in steps (the snap spacing). |
| **ORTHO** | `F8` | Lines go only straight across or straight up and down. |
| **POLAR** | `F10` | Guides at set angles while you draw. |
| **OSNAP** | `F3` | Object snaps: the cursor locks onto ends, middles, centres and more. |
| **ISO** | `F5` | Isometric drafting, and which isoplane is current. |
| **LWT** | | Show lineweights (line thickness) on screen. |
| **DYN** | `F12` | Show prompts and values next to the cursor. |
| **MODEL** | | Shows Model or Paper space; on a layout, click to switch. |

## Several drawings at once

Every drawing opens in its own tab. `Ctrl+Tab` and `Ctrl+Shift+Tab` step through them, `Ctrl+F4` or `CLOSE` closes the current one, and the **Window** menu lists them all. Each drawing keeps its own view, selection, undo history, layers and settings. Copy (`Ctrl+C`) and Paste (`Ctrl+V`) work between drawings, bringing layers, styles and blocks along.

When you close a drawing its view is remembered; the next time you open it, it comes back zoomed where you left it.

CHAPTER 4

# Giving Commands

*The command line, coordinates, and drawing aids.*

## Typing commands

- Type a command or its alias and press `Enter` or `Space`: `LINE` or `L`, `CIRCLE` or `C`, `TRIM` or `TR`.
- Answer the prompts the same way. Options are shown in square brackets: `[Close/Undo]`. Type the capital letters of one (`C`, `U`) to choose it, or click it. Or press `Down` (the arrow key) for a menu of them at the cursor; pick one with the arrow keys and `Enter`, or click it.
- The value in angle brackets is the default: `<1.0000>`. Press `Enter` to accept it.
- `Enter` on an empty prompt repeats the last command. `Esc` cancels. While a command runs, right-click is `Enter`.
- Menus, the ribbon and toolbar icons type the command for you; the prompts are the same.

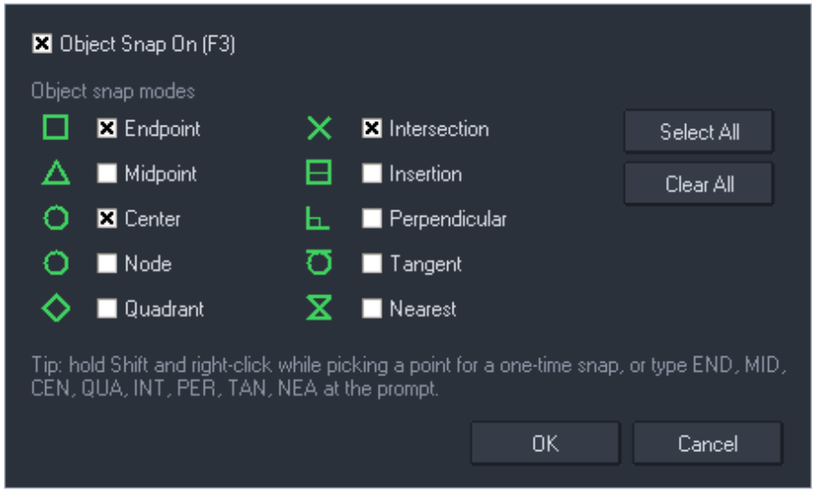
## Typing points and distances

| | |
|---|---|
| `3,4` | The point at X 3, Y 4 (the first point of a command). |
| `@2,0` | 2 to the right of the last point (relative). |
| `@5<45` | 5 away from the last point, at 45 degrees (polar). |
| `4,0` | After the first point, typed points are relative to the last one, as with AutoCAD's dynamic input: LINE `0,0` then `4,0` then `0,3` goes 4 across and 3 up. |
| `#10,10` | An absolute point, even after the first. |
| `5` | A bare number: 5 along the rubber band toward the cursor (direct distance). |
| `2'6 1 1/2` | Feet and inches, and fractions, wherever a distance is asked for. |

`SETVAR DYNPICOORDS 1` makes every typed point absolute (remembered). Points given by AutoLISP and scripts are always absolute.

## Object snaps

Object snaps put the point exactly on an object: its end, its middle, a circle's centre, where two lines cross. With **OSNAP** on (`F3`), the snap markers appear as the cursor nears an object; click to take the point. `OSNAP` (or right-click the OSNAP button) chooses which snaps are on.




*Object Snap settings (OSNAP).*

For one point only, type the snap's short name at the prompt: `END`, `MID`, `CEN`, `INT`, `PER`, `TAN`, `QUA`, `NOD`, `INS`, `NEA` - or `NON` for none. **Shift + right-click** gives a menu of them.

Snaps work on solids in every view, plan or 3D: the corners, edge middles and round edges' centres, and also their **faces**. **CEN** over a flat face snaps to the middle of that face; **NEA** with no edge near puts the point on the face under the cursor, so you can draw right onto a solid. In the 3D views **INT** finds where edges really meet in space, and where a line passes through a face; **PER** finds the foot of the perpendicular anywhere along the edge under the cursor.

## Ortho, Polar, Grid and Snap

`F8` ORTHO keeps lines straight across or up and down. `F7` GRID shows a grid and `F9` SNAP makes the cursor jump in steps. In the isometric views Ortho follows the X and Y of the plane you are drawing on; in Front, Back, Left and Right it follows the screen, so up the screen is up in Z.

## AutoLISP at the command line

Type AutoLISP straight in: `(+ 1 2)`, `!myvar` to see a variable, or a longer expression over several lines (the prompt shows `(_>` until the brackets balance). Expressions also work inside prompts: at "Specify next point" you can type `(polar (getvar "lastpoint") 0 5)`.

## Undo and redo

`U` or `Ctrl+Z` undoes the last command, `REDO` or `Ctrl+Y` brings it back, and `UNDO` undoes several at once. Each drawing has its own undo history.

CHAPTER 5

# Selecting, Grips and Properties

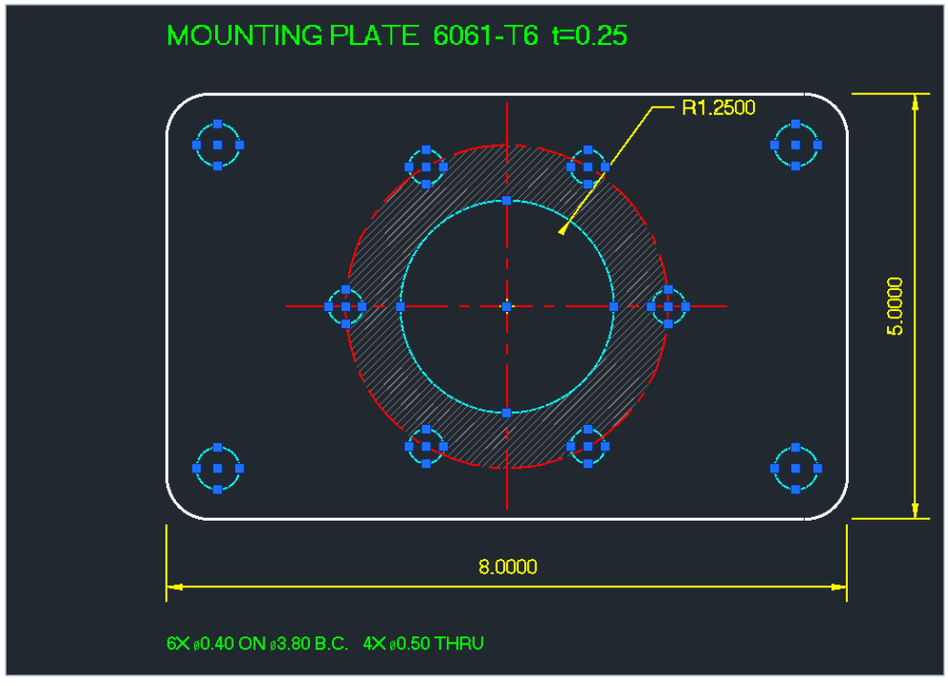
*Pick objects, change them by dragging, and read or edit everything about them.*

## Selecting

- **Click** an object at the `Command:` prompt to select it. Click more objects to add them.
- **Drag left to right** for a window: it selects what is completely inside.
- **Drag right to left** for a crossing window: it selects what is inside or touches it.
- `Ctrl+A` (SELECTALL) selects everything. `Esc` clears the selection.
- Then run a command - MOVE, ERASE (or press `Delete`) - and it uses the selection. Or start the command first and select when it asks.

## Grips

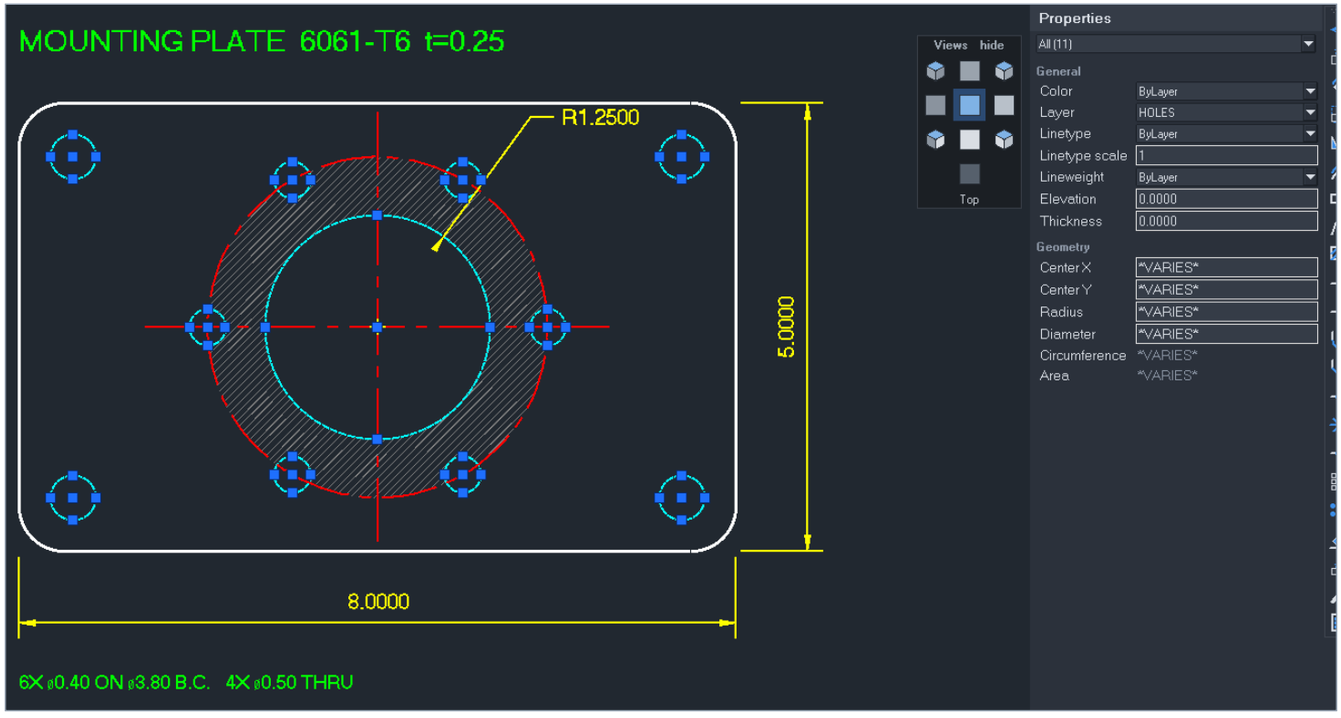
Selected objects show blue squares - **grips** - at their ends, middles and centres. Click a grip to make it hot (red) and drag it: a line's end moves, a circle grows. With a hot grip, `Enter` or `Space` steps through STRETCH, MOVE, ROTATE, SCALE and MIRROR, with the Base point, Copy, Undo and eXit options. Grips on several objects that share a point move them all together.




*Six bolt holes selected: their grips show at the centres and quadrants.*

## The Properties palette

`PROPERTIES` (`Ctrl+1`) opens the palette on the right. With nothing selected it shows the settings new objects will get (color, layer, linetype, lineweight). With a selection it shows what the objects share, and for one kind of object its sizes too - a circle's centre, radius, diameter, circumference and area. Change a value and press `Enter`. The list at the top ("All (6)", "Circle (6)") narrows a mixed selection to one kind.



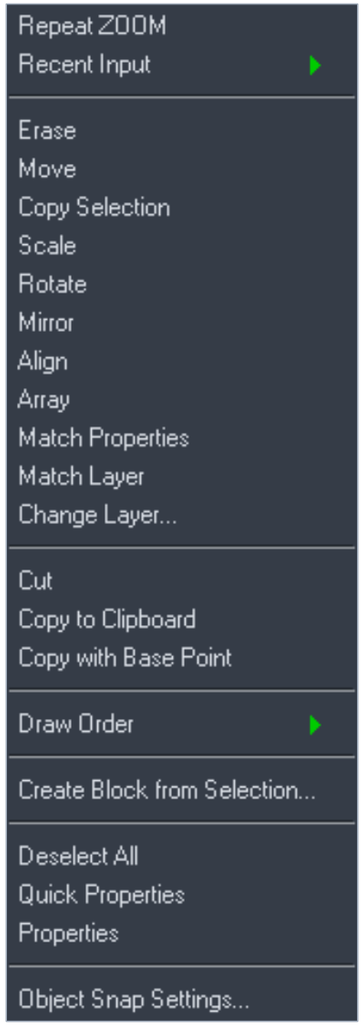

*The Properties palette showing the six selected circles.*

## Quick Properties and rollover tips

**Quick Properties** (`QP`, or right-click > Quick Properties) is a small panel with just the main properties; it stays open and follows whatever you select. Rest the cursor on any object for half a second and a **rollover tip** shows what it is and its sizes.

## The right-click menu

Right-click the drawing at the Command prompt: you get Repeat, Recent Input and, with objects selected, Erase, Move, Copy, Scale, Rotate, Mirror, Align, Array, Draw Order, Solid Editing (Union, Subtract, Intersect, Slice) for solids, Create Block, Match Properties and Properties.




*Right-click with objects selected.*

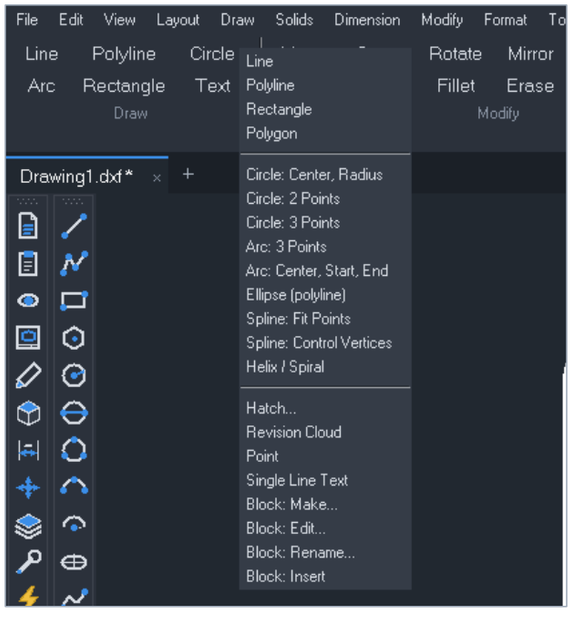
## Measuring

| | |
|---|---|
| `DIST` (`DI`) | Distance and angle between two points. |
| `ID` | The coordinates of a point. |
| `AREA` (`AA`) | Area and perimeter of points you pick or of an object. |
| `LIST` (`LI`) | Everything about the selected objects, in a window you can copy from. |
| `MASSPROP` | Area, centroid and moments of closed shapes; volume and mass of solids. |

CHAPTER 6

# Drawing

*Lines, curves, shapes and fills.*




*The Draw menu.*

## Lines and polylines

- **LINE** (`L`): click points; each pair is a separate line. `C` closes back to the start, `U` takes back a point, `Enter` ends.
- **PLINE** (`PL`): one object made of straight and curved segments. `A` switches to arcs, `L` back to lines, `W` sets a width.
- **RECTANG** (`REC`): two corners. Typing `11,8.5` after the first click makes an 11 x 8.5 rectangle.
- **POLYGON** (`POL`): number of sides, the centre, inscribed or circumscribed, and the size.

## Circles, arcs and ellipses

- **CIRCLE** (`C`): centre and radius, or `D` for a diameter; `2P`, `3P` and Tan-Tan-Radius.
- **ARC** (`A`): three points, or Start-Center-End and the other AutoCAD ways.
- **ELLIPSE** (`EL`): axis endpoints, or the centre; `Isocircle` in isometric drafting.

## Splines and helixes

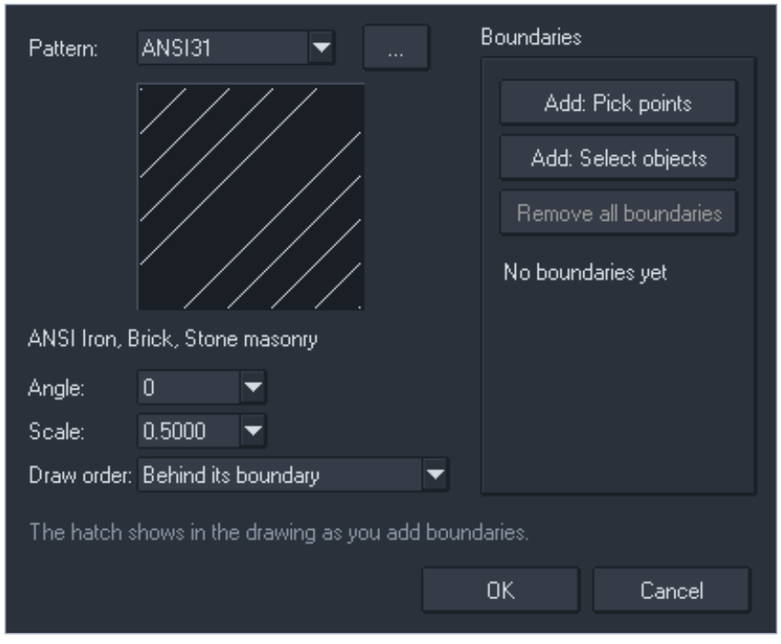
**SPLINE** (`SPL`): pick points and press `Enter`. With **Method Fit** the curve passes through every point; **Method CV** treats them as control vertices that pull the curve. Close, start and end Tangency, and Knots shape it. **HELIX** makes springs, thread paths and flat spirals: base centre, base and top radius, then height, with Turns, turn Height and tWist.

## Points, revision clouds and text

**POINT** (`PO`) places a point in the style set by **DDPTYPE** (Format > Point Style). **REVCLOUD** (`RC`) draws a cloud around changes. Text has its own chapter.

## Hatching

**HATCH** (`H`) fills an area. In the dialog, choose a pattern (or click the swatch for the palette), the angle and the scale, then **Add: Pick points** and click inside the area: the boundary is found automatically, with closed objects inside left as islands. The hatch shows as you go; `Enter` brings the dialog back, and **OK** keeps it.




*The Hatch dialog.*

Patterns include SOLID, ANSI31 to ANSI38, LINE, NET, DASH, DOTS, BRICK, SQUARE, CROSS, HONEY, EARTH, STEEL, INSUL and more. **HATCHEDIT** (`HE`) changes an existing hatch; **-HATCH** asks at the command line, for scripts.

CHAPTER 7

# Modifying

*Move, copy, trim, round off and rearrange.*

## Moving and copying

| | |
|---|---|
| **MOVE** (`M`) | Select, a base point, then where it goes. |
| **COPY** (`CO`) | The same, leaving the original; keep clicking for more copies. |
| **ROTATE** (`RO`) | A base point and an angle, or Reference to line one thing up with another. |
| **SCALE** (`SC`) | A base point and a factor, or Reference: "this length should be that length". |
| **MIRROR** (`MI`) | Two points on the mirror line; keep or erase the originals. |
| **ARRAY** (`AR`) | Copies in rows and columns (Rectangular) or around a centre (Polar). |
| **ALIGN** (`AL`) | Pairs of points: the source points move onto the destination points, turning and optionally scaling. Works in 3D with up to three pairs. |
| **STRETCH** (`S`) | A crossing window over the part to stretch, then a base point and where it goes. |

## Trimming and extending

**TRIM** (`TR`): select the cutting edges (or press `Enter` for all), then click the parts to cut off. **EXTEND** (`EX`) is the same, lengthening objects to the edges. `Shift`+click switches between them. They work on lines, arcs, circles and polylines. **BREAK** (`BR`) cuts a gap or splits an object at a point.

## Offset, fillet and chamfer

- **OFFSET** (`O`): a distance, then the object and the side - a parallel copy.
- **FILLET** (`F`): two lines (or a polyline with `P`) joined by an arc of the fillet radius (`R`). Radius 0 makes a sharp corner.
- **CHAMFER** (`CHA`): the same with a straight cut.
- **JOIN** (`J`) joins touching pieces into one; **EXPLODE** (`X`) breaks polylines, blocks, dimensions, text and solids into their parts; **PEDIT** (`PE`) edits polylines.

## Changing properties

- **MATCHPROP** (`MA`): pick a source object, then the objects to make like it. `S` chooses what is copied: color, layer, linetype, lineweight, text and dimension style, hatch pattern...
- **CHPROP**, the Properties palette, and the ribbon's Color, Linetype and Lineweight lists change selected objects directly.

## Draw order and text case

Later objects are drawn over earlier ones. **DRAWORDER** (`DR`) sends selected objects to the Front or Back, or Above or Under others; **TEXTTOFRONT** and **HATCHTOBACK** do it for all text or all hatches. **TCASE** changes text to UPPER, lower, Sentence, Title or tOGGLE case.

## Erasing and getting things back

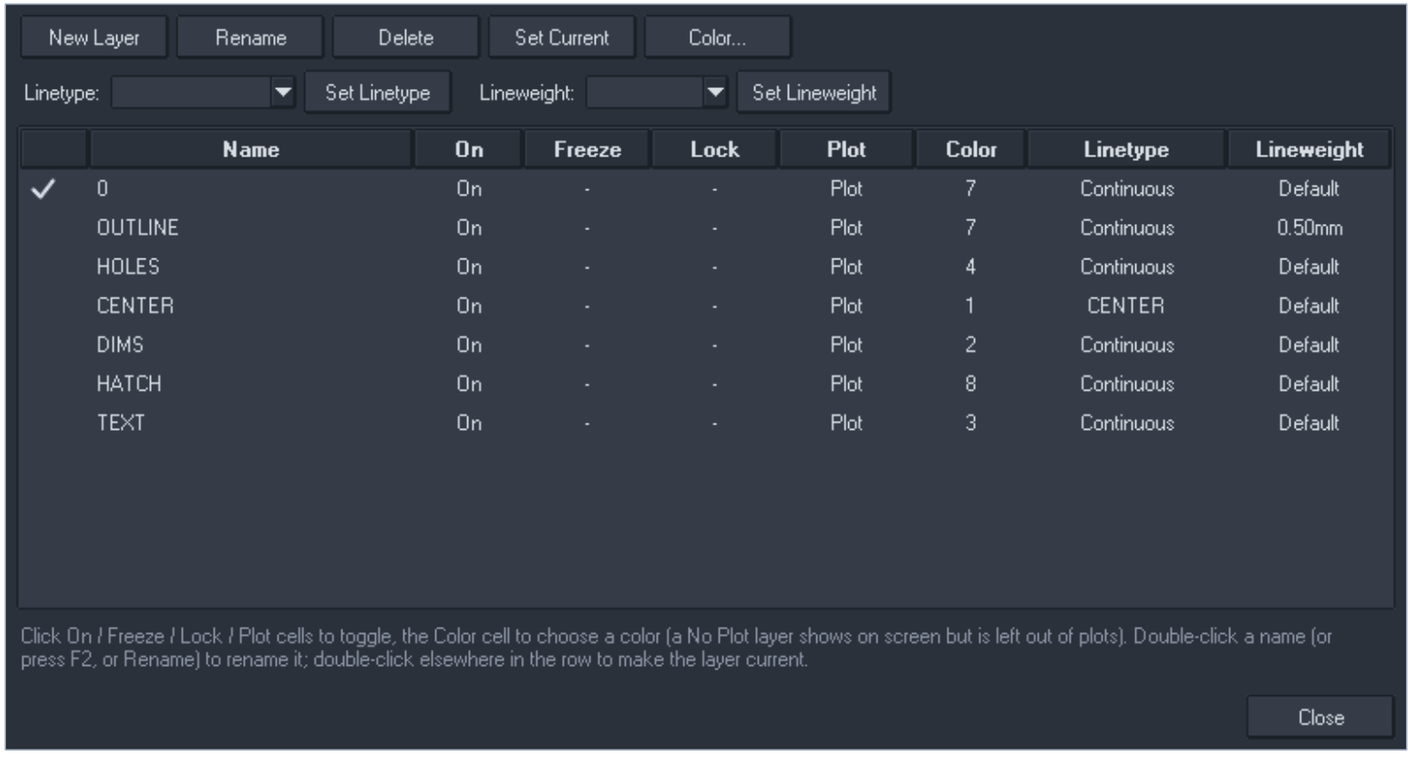
**ERASE** (`E`, or the `Delete` key) removes the selection; **OOPS** brings back the last erased objects; `U` undoes any command.

CHAPTER 8

# Layers, Colors and Linetypes

*Keep a drawing organised and make it print the way you want.*

Layers are like sheets of clear film stacked on each other: walls on one, dimensions on another, notes on a third. Each layer has a color, linetype and lineweight that its objects use, and can be turned off, frozen or locked.




*The Layer Properties Manager (LAYER).*

## The Layer Properties Manager

- Open it with **LAYER** (LA), **Format > Layer**, or Layer Properties on the ribbon.
- **New Layer** adds one with its name ready to type. Double-click a name (or F2) to rename it.
- Click **On**, **Freeze**, **Lock** or **Plot** in a row to switch it; click **Color**, **Linetype** or **Lineweight** to change them.
- Double-click a row (or **Set Current**) to make it the current layer - new objects go on it.
- A **No Plot** layer shows on screen but is left out of plots, like Defpoints.

## Quick layer tools

| | |
|---|---|
| **Make Current** (`LAYMCUR`) | Click an object: its layer becomes the current layer. |
| **Match** (`LAYMCH`) | Move the selected objects to the layer of an object you click. |
| **Change** (`CL`) | Move the selected objects to a layer picked from a list. |
| `LAYCUR` | Move the selected objects to the current layer. |
| `LAYISO` / `LAYUNISO` | Show only the layers of the selected objects, and back. |
| `LAYOFF` `LAYFRZ` `LAYLCK` | Turn off, freeze or lock an object's layer by clicking it (and LAYON, LAYTHW, LAYULK to undo). |

Choosing a layer in the ribbon's layer list while objects are selected moves them to it; with nothing selected it sets the current layer.

## Colors

Colors open the **Select Color** dialog: the 255 AutoCAD index colors, the standard colors and grays, ByLayer (the layer's color) and ByBlock. White objects (color 7) show black on a light background and print black.

## Linetypes and lineweights

**LINETYPE** (`LT`) loads CENTER, HIDDEN, DASHED, PHANTOM, DOT, DASHDOT, BORDER, DIVIDE and their variants, or any `.lin` file. **LTSCALE** (`LTS`) sets the size of the dashes for the whole drawing. **LWEIGHT** (`LW`) sets line thickness; turn on **LWT** in the status bar to see it.

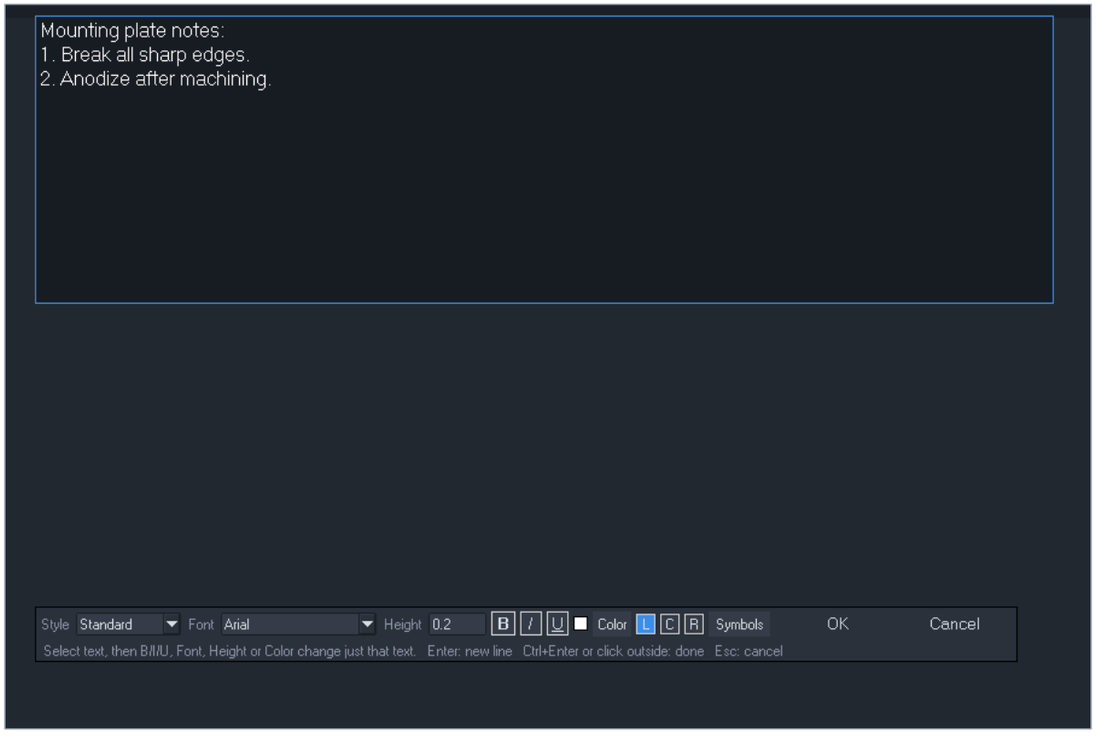
CHAPTER 9

# Text

*Notes, labels and title blocks, typed right on the drawing.*

## Paragraph text (MTEXT)

**MTEXT** (`T`, `MT`): pick two corners and an editor opens in that box. The text wraps at the box's width. `Enter` starts a new line; `Ctrl+Enter`, OK, or a click outside finishes; `Esc` cancels.




*Typing a note with MTEXT: the formatting bar is under the text box.*

The formatting bar has the text style, font, height, **Bold**, Italic and Underline (`Ctrl+B`, `Ctrl+I`, `Ctrl+U`), color, alignment, and a Symbols menu (degree, plus-minus, diameter...). Font, height, bold, italic, underline and color change only the selected characters. Drag the grips on the top corners of a selected MTEXT to change its width.

## Single-line text (TEXT)

**TEXT** (`DT`): pick the start point, the height and the angle, then type; each line is its own object. `Enter` on an empty line ends. Double-click any text (or **TEXTEDIT**, `ED`) to edit it.

## Text styles and fonts

**STYLE** (`ST`, Format > Text Style) sets a style's font, bold, italic, fixed height, width factor and oblique angle. Any installed font works, and AutoCAD `.shx` font names are shown with a similar font. To add fonts, put `.ttf` or `.otf` files in `~/.config/fancypencil/fonts`; they are installed the next time Fancy Pencil starts. Drawings that use Century Gothic show it when it is installed.

## Spelling and case

**SPELL** (`SP`) checks text, MTEXT, attributes and dimension text, with Change, Change All, Ignore and Add to Dictionary. While typing, unknown words are underlined in red; right-click one for suggestions. `Shift+F3` in the editor changes the case of the selected words.

## Text into outlines (TXTEXP)

**TXTEXP** (Express > Explode Text) turns text into closed polylines that trace each letter, in the same font you see on the screen, with the capitals as tall as the text height. Use it for lettering to cut, engrave or extrude: **EXTRUDE** the outlines to make raised letters. **EXPLODE** (`X`) does the same to TEXT; on MTEXT it first makes one TEXT for each line, so explode twice to get outlines. The outlines keep the text's layer and color. Undo puts the text back.




*Above, ordinary text; below, the same text after TXTEXP.*

CHAPTER 10

# Dimensions and Leaders

*Measurements and callouts that look like AutoCAD's.*

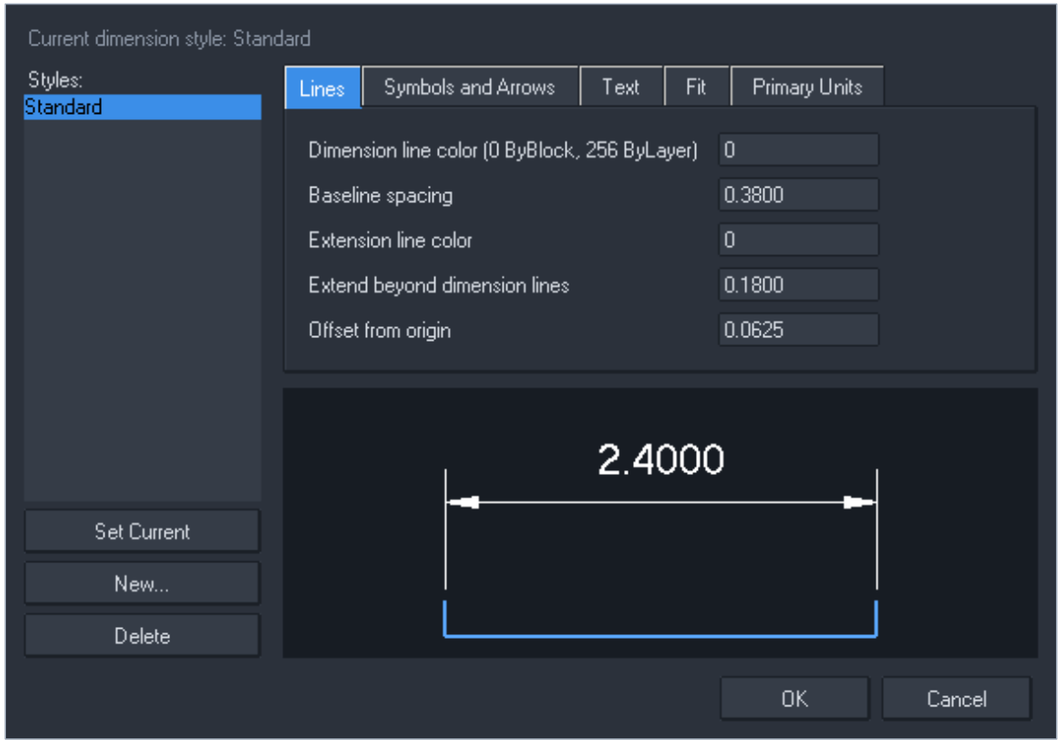
## Dimension commands

| | |
|---|---|
| **DIMLINEAR** (`DLI`) | Horizontal or vertical: two extension-line origins (or `Enter` and pick an object), then where the dimension line goes. |
| **DIMALIGNED** (`DAL`) | Parallel to the two points. |
| **DIMRADIUS** (`DRA`) | A circle's or arc's radius. |
| **DIMDIAMETER** (`DDI`) | A circle's diameter. |
| **DIMANGULAR** (`DAN`) | The angle between two lines, or of an arc. |
| **DIMCONTINUE** (`DCO`) | A chain of dimensions from the last one. |
| **DIMBASELINE** (`DBA`) | Dimensions stacked from the same first point. |

Bring a new dimension line near an existing parallel dimension and it snaps to the same line, so rows of dimensions line up. Grips at the arrows move the dimension line.

## Dimension styles

**DIMSTYLE** (`D`) opens the Dimension Style Manager: styles on the left (Set Current, New, Delete), and tabs for Lines, Symbols and Arrows, Text, Fit and Primary Units, with a live preview. Arrowheads include closed filled, open, dot, architectural tick and oblique. Changing a style updates every dimension that uses it.




*The Dimension Style Manager.*

## Leaders and multileaders

**LEADER** (`LE`): click the arrow tip, then the points of the line, press `Enter` and type the note. **Spline** or **STraight**, **Arrow** or **None**, and **NOText** are options at every point prompt. **MLEADER** (`MLD`) makes AutoCAD's newer multileader, whose arrow, line, landing and note are one object. Arrow size follows `DIMASZ` and the dimension style.

CHAPTER 11

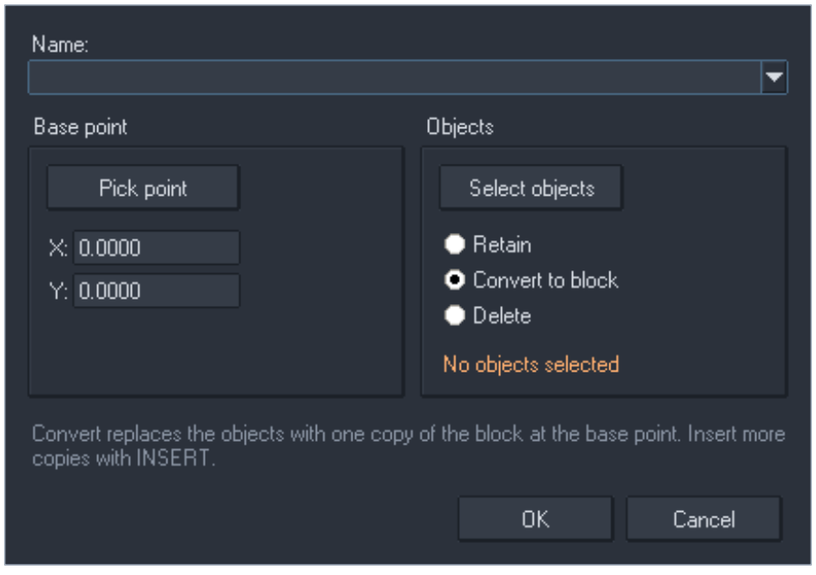
# Blocks

*Draw a part once, then use it again and again.*

A block is a group of objects saved under a name - a bolt, a door, a title block. Every copy (insert) of it changes when the block is edited.

## Making a block

**BLOCK** (`B`, Draw > Block: Make): type a name, **Pick point** for the base point (the point you will insert it by), **Select objects**, and choose whether the objects are retained, converted to the block, or deleted. Objects selected beforehand are used at once.




*The Block Definition dialog.*

## Inserting, editing and renaming

- **INSERT** (`I`): pick the block, the insertion point, scale and rotation.
- **BEDIT** (`BE`, or right-click a block > Edit Block...): the block opens in its own Block Editor tab with its base point at the red cross. Change it, then **BSAVE** (or `Ctrl+S`) and every copy updates. **BCLOSE** finishes.
- **RENAME** renames a block, layer, text style or dimension style everywhere it is used.
- **EXPLODE** (`X`) turns one insert back into separate objects; **BURST** keeps attribute values as text.
- Right-click a selection > **Create Block from Selection...**; Edit > **Paste as Block** (`Ctrl+Shift+V`).

A block made from objects at different heights keeps its 3D shape when inserted.

CHAPTER 12

# Views and 3D Navigation

*Zoom, pan, turn the drawing around and look at it from any side.*

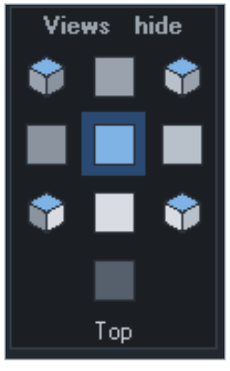
## Zoom and pan

- Turn the mouse **wheel** to zoom at the cursor; **middle-drag** to pan; **middle double-click** to see everything.
- **ZOOM** (Z): Extents (E), All, Object (0: the objects you pick), Window (W), Previous (P), Center, or a scale factor. **PAN** (P).

## When the drawing looks like a speck

If Zoom Extents shows your work as a dot, something in the drawing is far away from the rest, or far too big: a 3D move that went a hundred million units up, or a circle miles across. **STRAYS** (Tools > Find Stray Objects) finds them. It lists the drawing's groups of objects, with their size and where they are, and any very large objects. Pick a group to select it and zoom to it; then **MOVE** puts it back, or **ERASE** removes it. To keep a large object but out of the way, put it on a layer of its own and freeze that layer.

## The view picker and standard views




*The view picker.*

The **view picker** at the top right shows a cube for each standard view: Top in the middle, the Front, Back, Left and Right views around it, the four isometric views in the corners, and Bottom underneath. Click one to switch. **-VIEW** and **PLAN** do the same at the command line.

In every view you can draw and edit: Front, Back, Left, Right and Bottom each work like an AutoCAD UCS, with X to the right, Y up and Z toward you. A rectangle drawn in the Front view stands upright.

## Orbit

| | |
|---|---|
| **3DORBIT** (`3DO`) | Drag to turn the view around the target; up stays up. |
| **3DFORBIT** | Free orbit: turn any way inside the green circle, roll outside it. |
| **3DCORBIT** | Drag and let go: the view keeps turning. Click to stop. |
| **3DDISTANCE**, **3DSWIVEL** | Move the camera closer or away; turn the camera like turning your head. |
| `Shift` + middle-drag | Orbit at any time, even in the middle of a command. |

## Visual styles

**VSCURRENT** (`VS`, View > Visual Styles): **Wireframe** shows every edge, **Hidden** hides what is behind solids, **Shaded** fills and lights them, and **Shaded with Edges** adds the edges. Plots follow the visual style.

## Working planes

**UCS** sets any plane as the working plane: an object's or face's plane, three points, a new origin, or turned about X, Y or Z. **DRAFTPLANE** (`F6`) changes the plane you draw on in a 3D view without changing the view: Ground, Front, Right, three points, or an object's plane.

## Viewports on the Model tab

**VPORTS** splits the drawing area into 2, 3 or 4 viewports, each with its own view and style; **4 > Standard** gives Top, SE Isometric, Front and Right, lined up. Click a viewport to make it active (blue border); **Single** goes back to one.

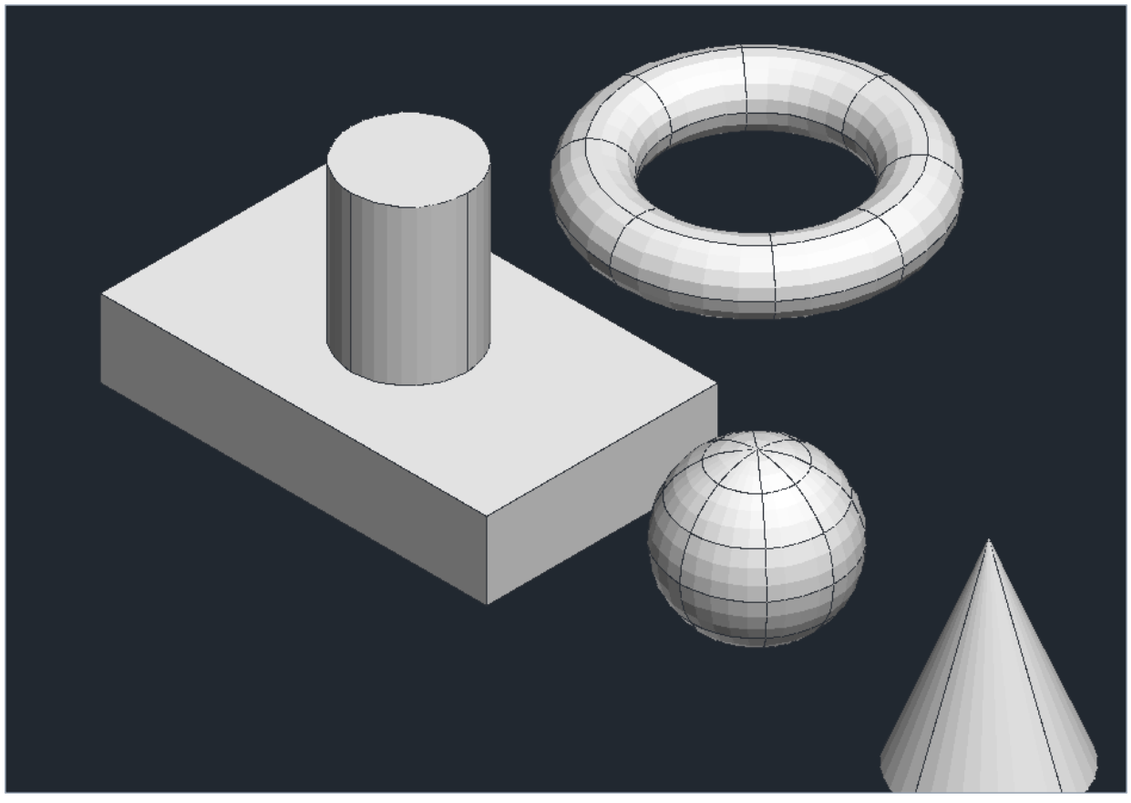
## Isometric drafting

**ISODRAFT** (the ISO button) turns on AutoCAD's 2D isometric mode: an isometric grid, and crosshairs and Ortho that follow the isoplane. `F5` switches between the Left, Top and Right isoplanes; ELLIPSE Isocircle draws a circle as it looks on the current one.

CHAPTER 13

# 3D Solids

*Build solid models, join and cut them, and print them in 3D.*



*Box, cylinder, sphere, cone and torus in the SE Isometric view, Shaded with Edges.*

## Basic solids

**BOX**, **WEDGE**, **CYLINDER**, **CONE**, **SPHERE** and **TORUS** (Solids menu) are drawn in the working plane, so a box drawn in the Front view stands upright. Heights may be negative.

## Solids from shapes

| | |
|---|---|
| **EXTRUDE** | Pulls a closed shape (circle, closed polyline, ellipse, closed spline) into a solid: by a height, a direction, or along a path. |
| **REVOLVE** | Turns a shape about an axis by any angle. |
| **SWEEP** | Carries a shape along a path - a line, arc, polyline, spline or helix. |

| | |
|---|---|
| **LOFT** | A solid through cross sections picked in order (a circle to a square, say). |
| **PRESSPULL** | Click inside a closed area, or on a flat face of a solid, and give a height. On a face, a positive height adds; a negative one cuts in. |

## Joining and cutting

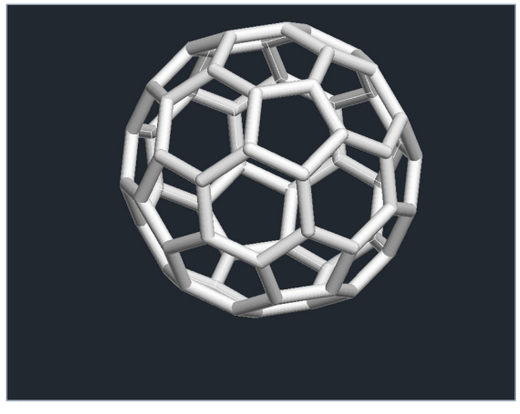
**UNION** joins solids into one. **SUBTRACT** removes some solids from others: pick what to keep, `Enter`, then what to remove. **INTERSECT** keeps only the shared part. **SLICE** cuts with a plane, keeping one side or both, and closes the cut with a flat face (left open where an imported mesh has gaps). All four are on the right-click menu under **Solid Editing** when solids are selected.

## Taking a solid apart

**EXPLODE** (`X`) splits a solid into its faces, as AutoCAD does: each flat face becomes a **region** and each curved one a **surface**. A box gives six regions; a cylinder two round regions and one surface. The faces keep the solid's layer and color, so you can move one aside, or erase one to look inside. Explode a region or surface again to get its edges as lines. Undo puts the solid back.

## Platonic solids, the soccer ball and struts

**TETRAHEDRON**, **OCTAHEDRON**, **DODECAHEDRON**, **ICOSAHEDRON** and **SOCCERBALL** (Solids > Platonic Solids & Soccer Ball): give the centre and the edge length. Type `W` for a wireframe (3D lines you can snap to), or `F` for a **Frame** of solid struts along every edge.



*A soccer ball frame: round struts with a ball at every corner, Shaded.*

**STRUTS** (`WIRESOLID`) does the same for any wireframe you draw: choose the section - **Rectangle**, **Polygon**, **Circle**, or **PLine** (a closed polyline you drew) - and whether to put a ball at every corner. Curved polylines, arcs and circles get one strut that follows the curve. **Ends** sets how far the struts stop short of the wire ends (negative: past them).

## Measuring and printing

**MASSPROP** reports volume, centroid and moments. **3DPRINT** saves solids as STL, 3MF or OBJ for a 3D printer's slicer, checking that each is watertight; **STLIN** brings in STL or OBJ models.

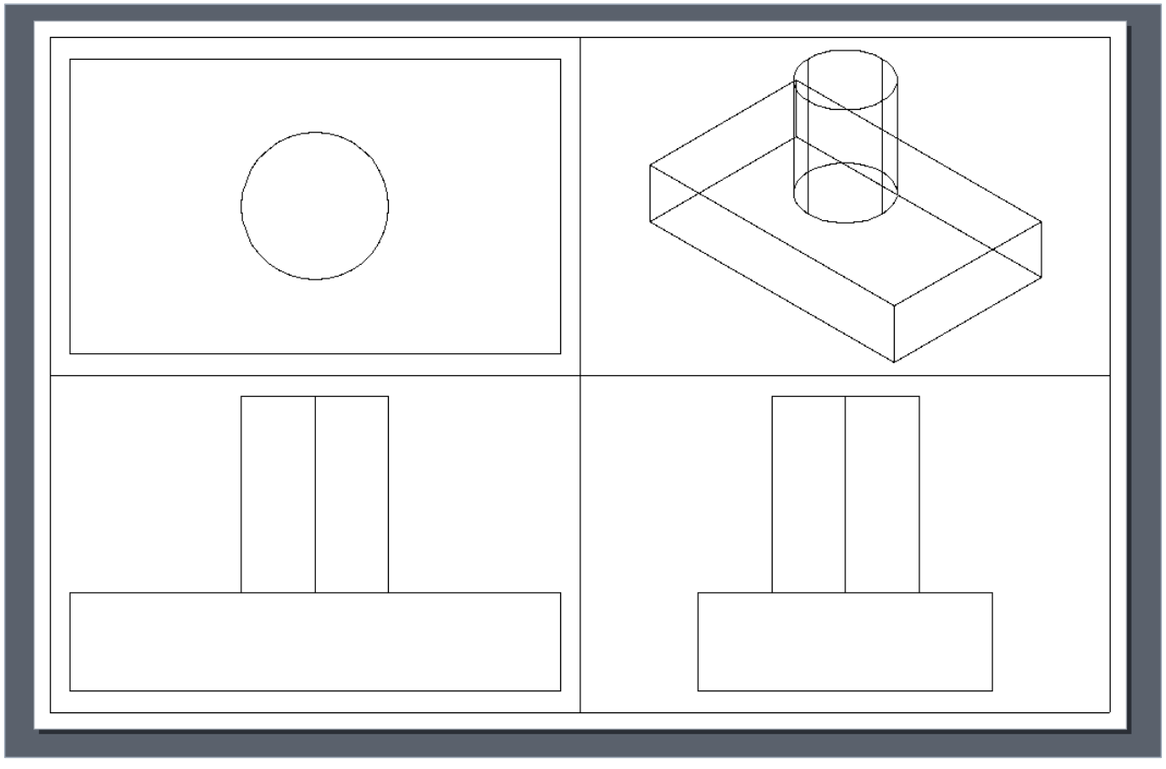
CHAPTER 14

# Layouts and Plotting

*Sheets of paper, viewports at scale, and printing to PDF or a printer.*

## Layouts

Below the drawing are the **Model** tab and layout tabs (**Layout1**, and more with **+**). A layout is a sheet of paper at its real size, with a dashed line for the printable area. Draw the border, title block and notes on it; they belong to that sheet.

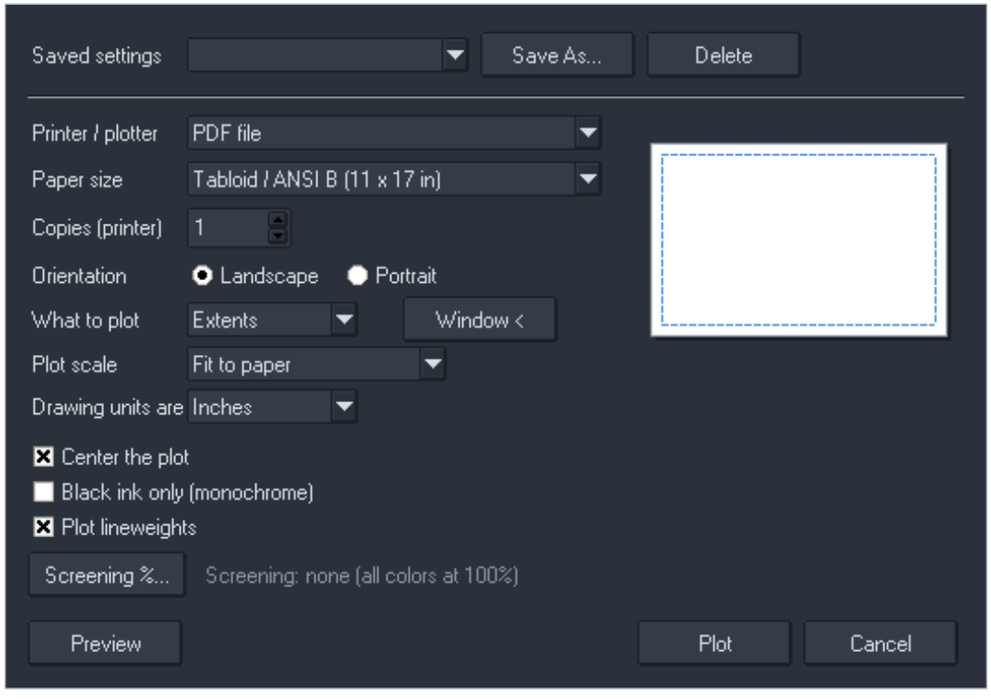


*A layout with four viewports (MVIEW 4, Standard): Top, Front, Right and SE Isometric.*

## Viewports on a layout

- **MVIEW** (`MV`): two corners, **Fit**, or **2**, **3** or **4** at once; **4 > Standard** makes Top, Front, Right and SE Isometric, lined up.
- **Double-click inside** a viewport (or `MSPACE`) to work through it: zoom, pan, change the view. **Double-click outside** (or `PSPACE`) to go back to the sheet.
- **VPSCALE**: 1:2, 1/4"=1'-0", 1:100, Fit... Properties shows and changes a selected viewport's scale, view and lock.
- **PAGESETUP** sets the sheet's paper size and orientation; right-click a layout tab to rename, copy or delete it.

## The Plot dialog




*PLOT (Ctrl+P).*

- **Printer/plotter**: "PDF file" or any printer your computer knows.
- **Paper size**: Letter, Legal, Tabloid, ANSI C-E, ARCH A-E, ISO A4-A0; portrait or landscape.
- **What to plot**: Display, Extents, Limits or Window. On a layout, the sheet at 1:1.
- **Plot scale**: Fit to paper, 1:1, 1:2 ... or architectural scales such as `1/4" = 1'-0"`.
- **Black ink only**, **Plot lineweights**, and **Screening %** to print colors lighter.
- **Saved settings**: Save As keeps everything under a name. Your last settings come back next time.
- **Preview** shows the page first.

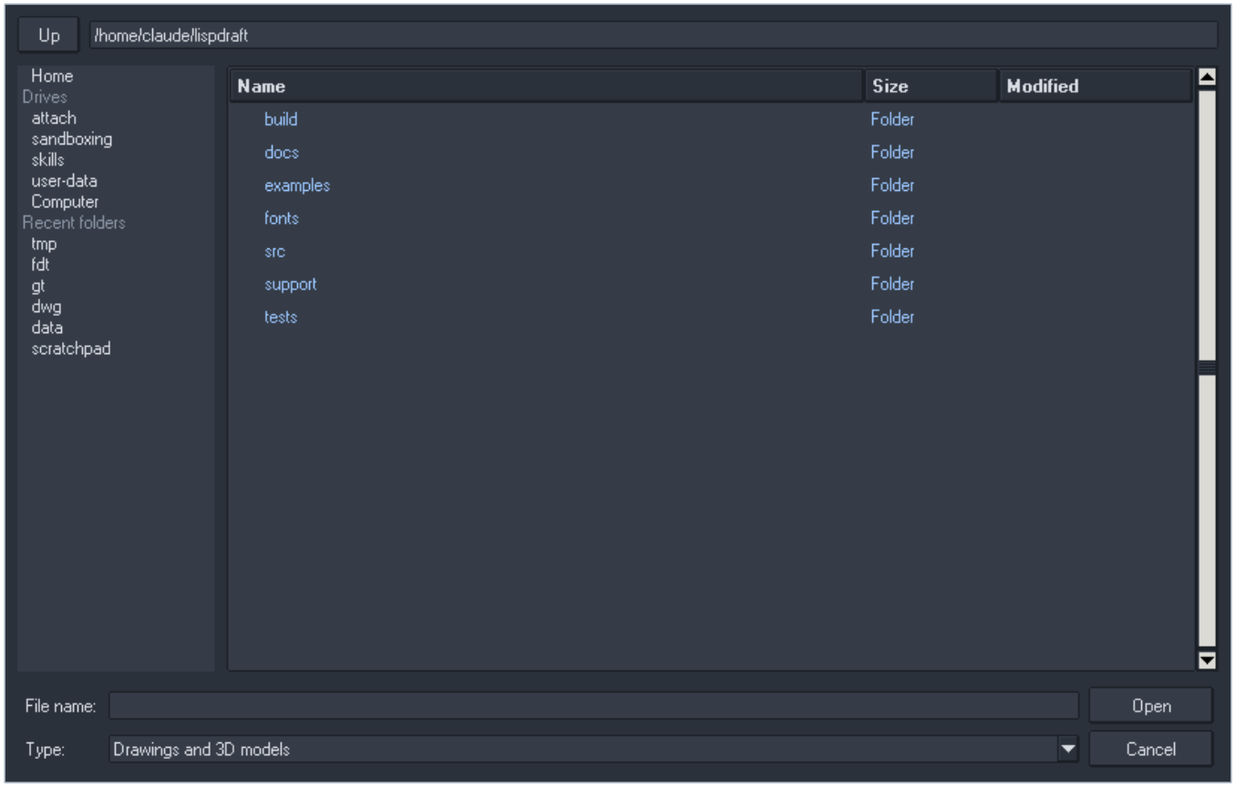
CHAPTER 15

# Files: Opening, Saving and Safety Copies

*DXF, DWG, pictures and PDFs - and what happens if the power goes.*

## Opening and saving

**OPEN** (`Ctrl+O`) shows Fancy Pencil's own file chooser: your folders, drives and recent folders on the left, the files with sizes and dates on the right. Type a path at the top and press `Enter` to go straight there. **File > Recent Files** lists the last 12 drawings.




*The Open dialog.*

**QSAVE** (`Ctrl+S`) saves; **SAVEAS** saves under a new name. Drawings are saved as AutoCAD 2000 DXF, which every AutoCAD since 2000 opens. **SAVEASR12** writes old R12 DXF. Saving over a drawing keeps the previous version beside it as a `.bak` file.

## DWG files

With ODA File Converter installed (Chapter 2), OPEN shows DWG files too, and Save writes them back as DWG (AutoCAD 2018 format). Without it, saving a DWG asks for a DXF name instead and never touches the DWG.

A DWG (or DXF) brings in its **raster images**, **PDF, DWF and DGN underlays** (looked for beside the drawing), **regions** and **3D solids**, **tables**, and its **paper-space layouts** with their viewports. A table comes in as a block showing its grid and text: edit it with **EXPLODE** and ordinary text editing. Anything Fancy Pencil can't keep is listed when the drawing opens.

## Safety copies and recovery

Every 10 minutes, each drawing with unsaved changes is copied to a safe place; your own file is not touched. If Fancy Pencil stops unexpectedly, the next start offers to open the copies. **File > Recover Safety Copies** opens them at any time, and `SAVETIME` sets the minutes between copies.

## Pictures, PDF pages and underlays

**IMAGEATTACH** places a JPEG, PNG, TIFF, GIF, BMP or WebP picture to trace over; **PDFATTACH** places a page of a PDF. Click the lower-left corner and give the width. **IMAGE** (the Image Manager) lists them, and Detach, Reload, Fade and Change Path. **DWFATTACH** and **DGNATTACH** attach DWF and MicroStation drawings as underlays. Keep the picture files with the drawing.

## 3D printing files

**3DPRINT** writes STL, 3MF or OBJ; **STLIN** (or File > Open on an STL or OBJ) reads them.

CHAPTER 16

# Toolbars and Your Settings

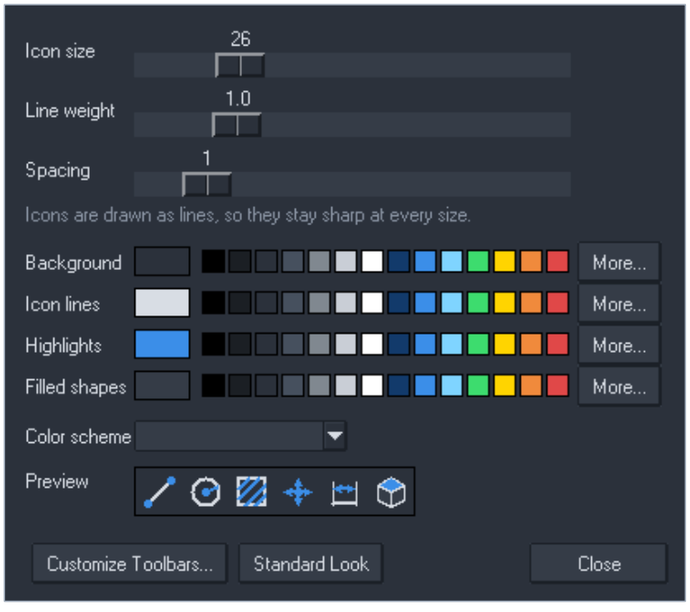
*Arrange the screen the way you like it.*

## Turning toolbars on and moving them

- **View > Toolbars**, or right-click any toolbar, to turn toolbars on and off. **Reset Toolbar Layout** brings back the standard one.
- Drag a toolbar by its dotted grip. Drop it near an edge to dock it there - top, bottom, left or right, even just outside the window's edge - or in the middle of the drawing to float it.
- A floating toolbar can also be dragged by its title bar: let go of it at an edge of the window and it docks there.
- **Clean Screen** (`Ctrl+0`, or View > Clean Screen) hides the ribbon and every toolbar so the drawing gets the whole window; `Ctrl+0` again brings them back.
- **Bring Floating Toolbars Back** (View > Toolbars) gathers every floating toolbar into the top left of the drawing, if any have wandered off or ended up behind other windows.

## Toolbar Options

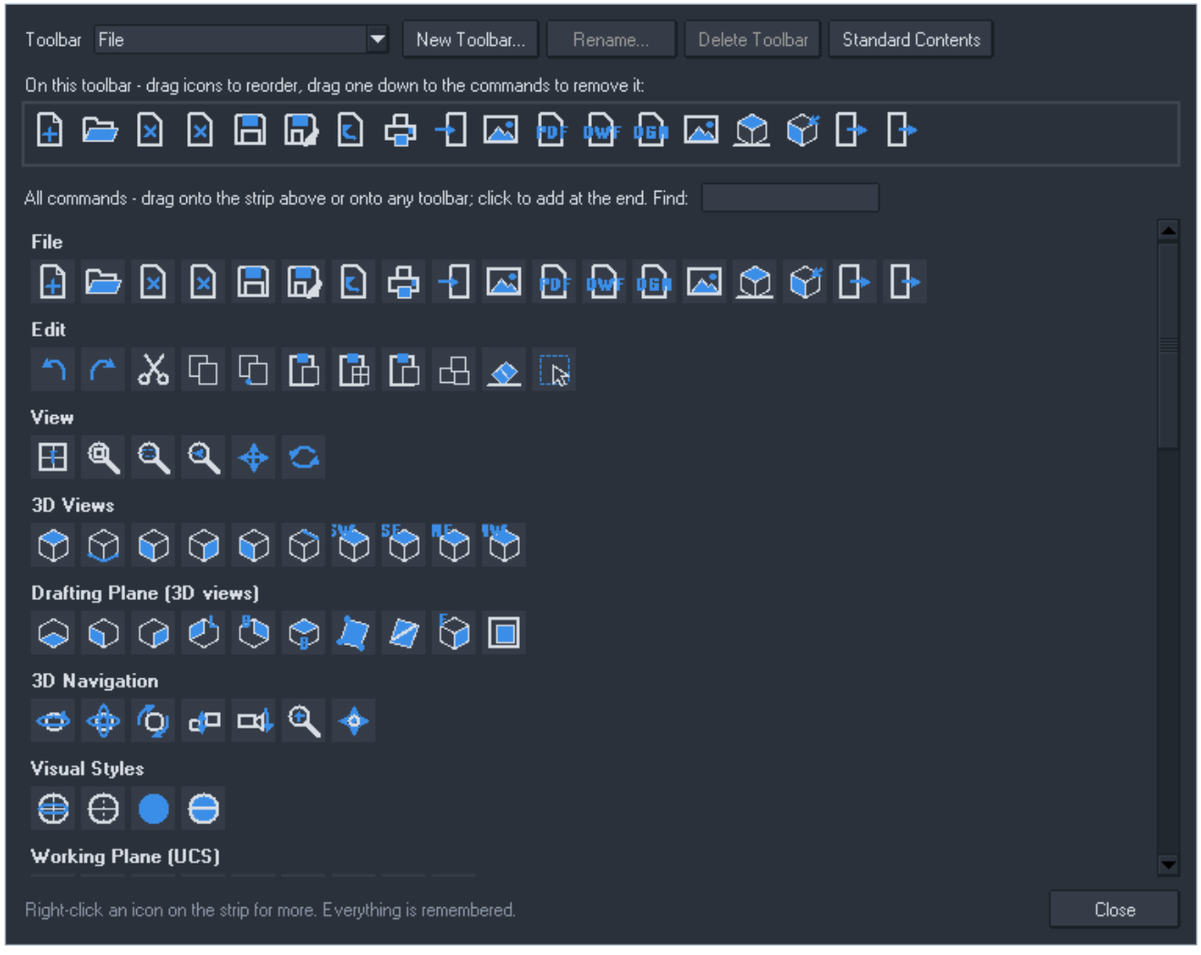
**Toolbar Options** (View > Toolbars, or `TOOLBAROPTIONS`) stays open while you work: sliders for icon size (16 to 64 pixels), line weight and spacing, colors for the background, icon lines, highlights and filled shapes, and ready-made color schemes. **Standard Look** puts everything back.




*Toolbar Options.*

## Customize Toolbars

**Customize Toolbars** is all drag and drop. At the top, the toolbar you are editing (or **New Toolbar...** for your own); below, every command's icon by menu, with a Find box. Drag a command onto the strip - or straight onto any toolbar on the screen. Drag icons along a toolbar to reorder them, or down onto the commands to remove them. At other times, `Shift`+drag moves a toolbar button.




*Customize Toolbars.*

## Command aliases

**Tools > Command Aliases > Edit Aliases...** (`ALIASEDIT`) changes the short names: `L` for LINE, your own for your AutoLISP commands. Your aliases are kept in `~/.config/fancypencil/fancypencil.pgp`, in AutoCAD's acad.pgp format.

## Cursor, selection and background

**OPTIONS** (`OP`, Tools > Cursor, Selection & Background): crosshair size and color, pickbox, snap marker and grip sizes, rollover tips, and the drawing background (dark, black, gray, paper, white or any color). These settings are yours: they apply in every drawing and are remembered.

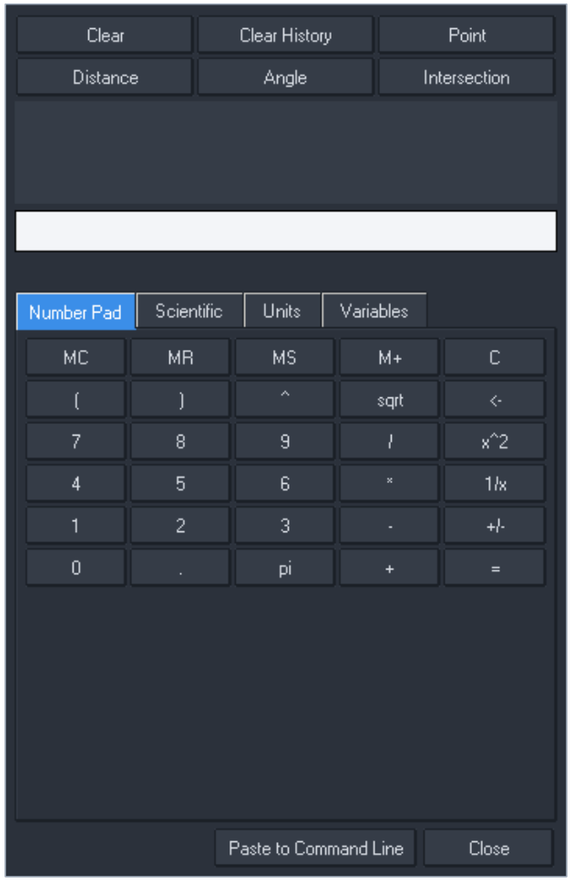
CHAPTER 17

# Tools

*Calculating, units, limits and the command history.*

## QuickCalc

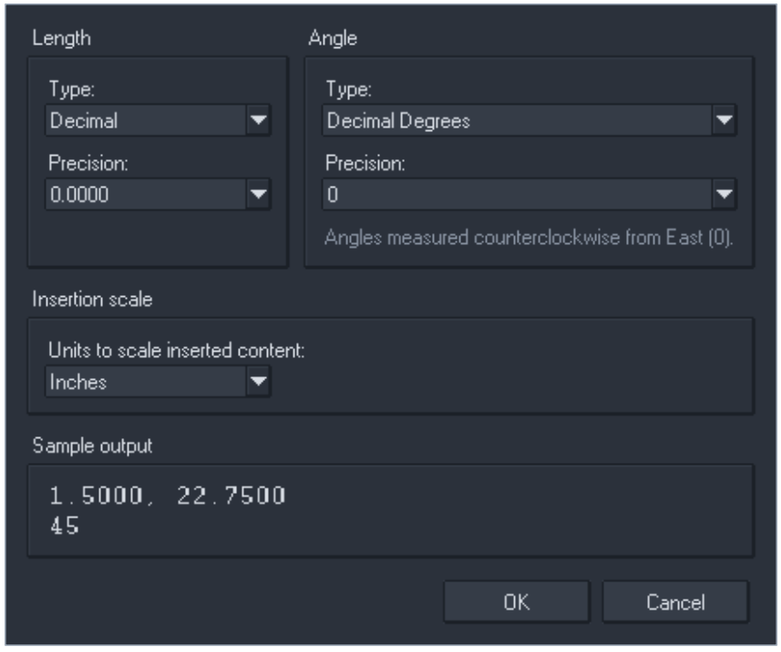
**QUICKCALC** (`QC`, `Ctrl+8`) opens a calculator beside the drawing. Type an expression and press `Enter`: `5'6"*2`, `sqrt(2)*3`, `dist(end,end)/2`. It has number and scientific keys, unit conversion and variables, and buttons that read a point, distance or angle from the drawing. In the middle of a command, type `'QC` at a prompt, work out the value, and **Apply** gives it to the command.




*QuickCalc.*

## Units

**UNITS** (Format > Units): how lengths are shown - Architectural (`12'-6 1/2"`), Decimal, Engineering, Fractional or Scientific - with their precision; angles in degrees, deg/min/sec, grads, radians or surveyor's units. The sample shows the result as you choose.




*Drawing Units.*

## Limits, point style and settings

- **LIMITS** (Format > Drawing Limits): the drawing's size, from corners or a sheet size and scale.
- **DDPTYPE** (Format > Point Style): the 20 point symbols, and their size.
- **SETVAR**, or type any system variable's name, to see or change it.

## Command history

**HISTORY** (F2) shows everything from the command line for the whole session, with Find, Copy All and Save As. Tick **Keep a log file** to have it saved every day.

CHAPTER 18

# AutoLISP

*Run AutoCAD's programming language, and your own routines.*

Fancy Pencil's AutoLISP follows AutoCAD's rules: `(defun c:name (args / locals) ...)` makes a command, symbols aren't case-sensitive, `(/ 7 2)` is 3, and `*error*` handlers work. About 250 functions are built in.

## Loading routines

- **APPLOAD** (`AP`, Tools > Load AutoLISP...) loads a `.lsp` file.
- `(load "myroutine.lsp")` at the command line does the same.
- Files named `acaddoc.lsp` or `fancypencil.lsp` in the current folder, and `~/.fancypencil.lsp`, load at startup, and `S::STARTUP` runs if defined.
- Give a routine an alias in your alias file: `HH, *HELLO` runs `C:HELLO`.

## An example

```
(defun c:box2 (/ p)
 (setq p (getpoint "\nCorner: "))
 (command "rectang" p (polar p 0 4) )
 (princ))
```

Type that at the command line (or put it in a file and load it), then `BOX2`.

## What is built in

| | |
|---|---|
| **Core** | car/cdr, cons, list, append, nth, member, assoc, mapcar, apply, lambda, foreach, repeat, while, cond, if, setq, defun |
| **Math** | + - * / rem abs fix float max min sqrt expt exp log sin cos atan logand |
| **Strings** | strcat strlen substr strcase itoa atoi atof rtos distof angtos chr ascii wcmatch, vl-string-* |
| **Geometry** | distance angle polar inters trans osnap textbox |
| **Input** | getpoint getcorner getdist getangle getreal getint getstring getkword initget entsel grread getfiled alert |
| **Drawing** | entget entmod entmake entdel entlast entnext handent tblsearch tblnext ssget ssadd sslength ssname sssetfirst |
| **Commands** | command, command-s, vl-cmdf, with pause |

| | |
|---|---|
| **Visual LISP** | vl-sort vl-remove-if vl-position vl-some vl-every vl-catch-all-apply ... |
| **Express Tools** | acet-error-init, acet-str-*, acet-ss-*, acet-geom-*, acet-pline-make ... |

`ssget` supports interactive selection, "X", "W", "C", "WP", "CP", "F", "L", "P", "I" and ":S" with filter lists, wildcards and `-4` operators. Not available: ActiveX (`vla-*`, `vlax-*`) and DCL dialogs.

## Examples that come with Fancy Pencil

`/opt/fancypencil/examples/demo-plate.lsp` draws the dimensioned plate on page 1 (load it and type `PLATE`). `tools.lsp` has TLEN (total length), BCIRC (bolt circle), NUM (numbering) and CL (centre marks).

CHAPTER 19

# Keyboard Shortcuts

*The keys that save the most time.*

| Key | Command | What it does |
|---|---|---|
| `Enter` `Space` | | Finish typing; repeat the last command at an empty prompt. |
| `Esc` | | Cancel the command; clear the selection. |
| `Delete` | ERASE | Erase the selected objects. |
| `Ctrl+N` | NEW | New drawing. |
| `Ctrl+O` | OPEN | Open a drawing. |
| `Ctrl+S` | QSAVE | Save. |
| `Ctrl+P` | PLOT | Plot. |
| `Ctrl+Z` / `Ctrl+Y` | U / REDO | Undo / redo. |
| `Ctrl+C` / `Ctrl+X` | COPYCLIP / CUTCLIP | Copy / cut to the clipboard. |
| `Ctrl+V` | PASTECLIP | Paste at a point. |
| `Ctrl+Shift+C` | COPYBASE | Copy with a base point. |
| `Ctrl+Shift+V` | PASTEBLOCK | Paste as a block. |
| `Ctrl+A` | SELECTALL | Select everything. |
| `Ctrl+0` | CLEANSCREENON | Clean Screen: hide the ribbon and toolbars (again to show). |
| `Ctrl+1` | PROPERTIES | Properties palette. |
| `Ctrl+8` | QUICKCALC | Calculator. |
| `Ctrl+Tab` | | Next drawing tab (`Ctrl+Shift+Tab`: previous). |
| `Ctrl+F4` | CLOSE | Close the drawing. |
| `F1` | MANUAL | Open this manual. |
| `F2` | HISTORY | Command history. |
| `F3` | OSNAP | Object snaps on/off. |
| `F5` | ISOPLANE | Next isoplane (isometric drafting). |
| `F6` | DRAFTPLANE | Next drafting plane in 3D views. |

| Key | Command | What it does |
|---|---|---|
| `F7` | GRID | Grid on/off. |
| `F8` | ORTHO | Ortho on/off. |
| `F9` | SNAP | Snap on/off. |
| `F10` | POLAR | Polar tracking on/off. |
| `F12` | DYN | Dynamic input (prompts at the cursor) on/off. |
| Wheel | | Zoom at the cursor. |
| Middle-drag | PAN | Pan. |
| Middle double-click | ZOOM E | Zoom to everything. |
| `Shift`+middle-drag | 3DORBIT | Orbit, at any time. |
| `Down` arrow | | During a command: a menu of its options. At the Command prompt: earlier commands (with `Up`). |
| Right-click | | Menu at the Command prompt; Enter during a command. |
| `Shift`+right-click | | One-time object snap menu. |

CHAPTER 20

# Command Reference

*Every command, A to Z, with its aliases (the short names you can type instead).*

## 3D

**3DCORBIT** Continuous orbit: drag and let go, and the view keeps turning.

**3DDISTANCE** Moves the camera closer to the objects or farther away (drag up and down).

**3DFORBIT** Free orbit: turns the view in any direction, without keeping up up.

**3DORBIT** `3DO, ORBIT` Turns the view around a target point (constrained orbit: up stays up).

**3DPAN** Pans a 3D view (drag).

**3DPRINT** `EXPORT3D, PRINT3D` Saves solids as a file for a 3D printer's slicer: STL, 3MF or OBJ.

**3DSWIVEL** Turns the camera where it stands, like turning your head: the target moves.

**3DZOOM** Zooms in a 3D view (drag up to zoom in).

## A

**ABOUT** Shows program information.

**ALIASEDIT** `ALIASES` Edits command aliases.

**ALIASFILE** `PGP` Opens your command alias file (fancypencil.pgp) in a text editor.

**ALIASFOLDER** Opens the folder that holds your command alias file.

**ALIGN** `3DALIGN, AL` Moves, rotates and optionally scales objects so up to three points line up with three others, in 3D.

**APPLOAD** `AP` Loads an AutoLISP (.lsp) file.

**ARC** `A` Creates an arc.

**AREA** `AA` Calculates the area and perimeter of objects or a polygon of points.

**ARRAY** `AR` Creates multiple copies in a rectangular or polar pattern.

**ARRAYPOLAR** Polar array.

**ARRAYRECT** Rectangular array.

**ATTACH** Attaches a picture or a PDF page to the drawing (choose the file).

## B

**BACKGROUND** `BGCOLOR, BKG` Sets the drawing area's background color.

**BCLOSE** Closes the Block Editor.

**BEDIT** `BE, BLOCKEDIT, REFEDIT` Opens a block in the Block Editor to change it.

**BLOCK** `B, BMAKE` Creates a block definition from selected objects (Block Definition dialog).

**-BLOCK** Creates a block definition at the command line.

**BOX** Creates a 3D solid box in the working plane.

**BREAK** `BR` Breaks the selected object between two points.

**BREAKLINE** `BREAKL` Express: draws a break-line symbol.

**BRINGTOFRONT** Draws the selected objects over everything else.

**BSAVE** Saves the Block Editor's changes into the block.

**BURST** Express: explodes blocks, turning attribute definitions into text.

## C

**CAL** Calculates an expression (numbers, points, geometry picked in the drawing).

**CHAMFER** `CHA` Bevels the edges of objects.

**CHANGELAYER** `CL, LAYCHANGE` Moves selected objects to a layer you choose from a list.

**CHPROP** `CHANGE` Changes the layer or color of objects.

**CIRCLE** `C` Creates a circle.

**CLEANSCREENOFF** Brings the ribbon and toolbars back after Clean Screen (Ctrl+0).

**CLEANSCREENON** Clean Screen: hides the ribbon and all toolbars so the drawing has the whole window (Ctrl+0).

**CLEARRECENT** Empties the recent files list.

**CLOSE** Closes the current drawing.

**CLOSEALL** Closes all open drawings.

**COLOR** `COL, DDCOLOR` Sets the color for new objects (the Select Color dialog in the window).

**CONE** Creates a 3D solid cone (or a cut-off cone with a top radius).

**COPY** `CO, CP` Copies objects.

**COPYBASE** Copies objects to the clipboard with a base point you pick (Ctrl+Shift+C).

**COPYCLIP** Copies selected objects to the clipboard (Ctrl+C); paste them into any drawing.

**COPYM** Express: copies objects repeatedly, by repeat, divide or measure.

**CUTCLIP** Copies the selection to the clipboard and erases it (Ctrl+X).

**CYLINDER** `CYL` Creates a 3D solid cylinder standing on the working plane.

## D

**DDPTYPE** `POINTSTYLE, PTYPE` Point Style dialog: how points are shown and how big.

**DESELECT** Clears the current selection (grips).

**DGNATTACH** Attaches a MicroStation DGN file as an underlay.

**DIMALIGNED** `DAL, DIMALI` Creates an aligned linear dimension.

**DIMANGULAR** DAN, DIMANG Creates an angular dimension.

**DIMBASELINE** DBA, DIMBASE Creates a dimension from the baseline of the last one.

**DIMCONTINUE** DCO, DIMCONT Creates a dimension starting at the second extension line of the last one.

**DIMDIAMETER** DDI, DIMDIA Creates a diameter dimension for a circle or arc.

**DIMLINEAR** DIMLIN, DLI Creates a horizontal or vertical linear dimension.

**DIMRADIUS** DIMRAD, DRA Creates a radius dimension for a circle or arc.

**DIMREGEN** Redraws every dimension's picture from its points and current style.

**DIMSTYLE** D, DDIM, DIMSTY, DST Creates and restores dimension styles.

**DIST** DI Measures the distance and angle between two points.

**DIVIDE** DIV Places evenly spaced point objects along a line, arc or circle.

**DOCCLOSE** Closes an open drawing (used by the file tabs' close buttons).

**DOCNEXT** Switches to the next open drawing (Ctrl+Tab).

**DOCPREV** Switches to the previous open drawing (Ctrl+Shift+Tab).

**DOCSWITCH** Makes another open drawing current (used by the file tabs).

**DODECAHEDRON** DODECA Creates a dodecahedron: 12 pentagonal faces.

**DRAFTPLANE** DPLANE, DRP Chooses the plane you draw on in the 3D views, without changing the view: Ground, Front, Right, Left, Back, Bottom, through three points, on an object or a solid's face, facing the screen.

**DRAWINGRECOVERY** RECOVERCOPIES Opens safety copies left by Fancy Pencil stopping unexpectedly.

**DRAWORDER** DR Changes which objects are drawn on top: Front, Back, Above or Under other objects.

**DWFATTACH** Attaches a DWF or DWFx file as an underlay.

**DXFIN** IMPORT Imports a DXF drawing into the current drawing.

**DXFOUT** Exports the drawing as DXF (choose AutoCAD 2000 or R12).

## E

**ELEV** ELEVATION Sets the elevation and thickness for new objects.

**ELLIPSE** EL Creates an ellipse (and isocircles in isometric drafting).

**ERASE** DELETE, E Removes objects from the drawing.

**EXPLODE** X Breaks a compound object into its component objects.

**EXTEND** EX Extends objects to meet the edges of other objects.

**EXTRIM** Express: trims every object crossing a cutting edge on one side.

**EXTRUDE** EXT Extrudes closed shapes into solids (open ones into surfaces): by a height, along a direction, or along a path.

## F

**FILLET** F Rounds and fillets the edges of objects.

**FLATTEN** Express: sets the Z coordinate of selected objects to zero.

## G

**GETODA** ODA Opens a window that downloads and installs the free ODA File Converter (for .dwg files).

**GETSEL** Express: selects objects by layer and type.

**GRID** Turns the grid on or off, or sets spacing.

## H

**HATCH** BH, BHATCH, GRADIENT, H Fills an enclosed area with a hatch pattern or solid fill.

**HATCHEDIT** HE Changes the pattern, scale or angle of existing hatches.

**HATCHTOBACK** Sends every hatch and fill behind everything else.

**HELIX** Creates a helix (a spring or thread path) or a flat spiral.

**HELP** ? Lists commands.

**HIDE** HI Shows the drawing with hidden lines removed (visual style Hidden).

**HISTORY** HIS, TEXTSCR Shows the whole record of this session: every command, answer and message.

**HPSET** Sets the hatch pattern, scale and angle used by HATCH.

## I

**ICOSAHEDRON** ICOSA Creates an icosahedron: 20 triangular faces.

**ID** Displays the coordinates of a location.

**IMAGE** EXTERNALREFERENCES, IM, IMAGEMANAGER, XR, XREF The Image Manager: the pictures and PDF pages attached to the drawing.

**IMAGEATTACH** IAT Attaches a picture (PNG, JPEG, TIFF, GIF, BMP) to the drawing.

**INSERT** DDINSERT, I Inserts a block or a DXF drawing.

**INTERSECT** IN Keeps only the part that solids share.

**ISODRAFT** Turns isometric drafting on (choosing an isoplane) or off.

**ISOPLANE** Chooses the isoplane for isometric drafting (Left, Top or Right).

## J

**JOIN** J Joins lines, arcs and open polylines end to end into one polyline.

## K

**KEEPPROFILES** Keeps (On) or removes (Off) the circles, polylines and paths that EXTRUDE, REVOLVE, SWEEP and LOFT use (DELOBJ).

## L

**LAYCUR** Express: moves selected objects to the current layer.

**LAYDEL** Express: deletes a layer and every object on it.

**LAYER** DDLMODES, LA, LAYERP Manages layers.

**LAYFRZ** Express: freezes the layer of a selected object.

**LAYISO** Express: isolates the layers of selected objects (turns the others off).

**LAYLCK** Express: locks the layer of a selected object.

**LAYMCH** LMH, MATCHLAYER Changes selected objects to the layer of a destination object (Match Layer).

**LAYMCUR** LAYERMAKECURRENT, LMC Makes the layer of a selected object current.

**LAYMRG** Express: merges layers into a target layer.

**LAYOFF** Express: turns off the layer of a selected object.

**LAYON** Express: turns all layers on.

**LAYOUT** LO Makes, copies, deletes, renames and shows layouts (paper space sheets).

**LAYOUTCOPY** Copies a layout from its tab.

**LAYOUTDELETE** Deletes a layout from its tab.

**LAYOUTMOVE** Moves a layout tab one place left or right.

**LAYOUTNEW** Adds a layout (the + after the layout tabs).

**LAYOUTPAGESETUP** Page setup for a layout from its tab.

**LAYOUTRENAME** Renames a layout from its tab.

**LAYOUTTAB** Shows a layout tab by its number (0 = Model); used by the tabs under the drawing.

**LAYTHW** Express: thaws all layers.

**LAYULK** Express: unlocks the layer of a selected object.

**LAYUNISO** Express: restores layers turned off by LAYISO.

**LEADER** `LE, LEAD, QLEADER` Creates a leader: an arrow and a straight or spline line pointing at a note.

**LIMITS** `DRAWINGLIMITS` Sets the drawing's size (its limits): the area the grid, ZOOM All and Plot Limits use.

**LINE** `L` Creates straight line segments.

**LINETYPE** `DDLTYPE, LT, LTYPE` Loads, sets and lists linetypes.

**LIST** `LI, LS` Displays property data for selected objects.

**LOFT** Makes a solid (or surface) through cross sections picked in order.

**LOGFILEOFF** Stops writing the command line to the log file.

**LOGFILEON** Starts writing everything shown in the command line to a log file.

**LTSCALE** `LTS` Sets the global linetype scale factor.

**LWEIGHT** `LINEWEIGHT, LW` Sets the current lineweight and lineweight display.

## M

**MANUAL** `F1, USERMANUAL` Opens the Fancy Pencil Manual (a PDF) in your PDF viewer.

**MASSPROP** `MASSPROPERTIES` Mass properties of regions (closed flat objects) and solids: area, volume, centroid, moments of inertia ...

**MATCHPROP** `MA, PAINTER` Copies properties (layer, color, linetype, text style, dimension style, hatch ...) from one object to others.

**MIRROR** `MI` Creates a mirrored copy of objects.

**MLEADER** `MLD` Creates a multileader: an arrow, a landing and a note, as one object.

**MOCORO** Express: moves, copies, rotates and scales with one base point.

**MODEL** `TILEMODE1` Shows the Model tab.

**MOVE** `M` Moves objects.

**MPEDIT** Express Tools: edits several polylines at once.

**MSPACE** `MS` Works in model space through a layout viewport.

**MSPACETOGGLE** Switches between paper space and the last viewport (the MODEL/PAPER button).

**MTEXT** `MT, T` Creates multiline (paragraph) text, typed in place on the drawing.

**MULTIPLE-POINT** Creates points until Enter.

**MVIEW** `MV` Makes layout viewports: windows on the sheet that show model space.

## N

**NEW** `QNEW` Creates a new drawing in its own tab.

## O

**OCTAHEDRON** `OCTA` Creates an octahedron: 8 triangular faces.

**OFFSET** `O` Creates concentric circles, parallel lines, and parallel curves.

**OOPS** Restores the last erased objects.

**OPEN** Opens a drawing (DXF, or DWG through a converter), or a 3D model (STL, OBJ) as a solid.

**OPENRECENT** Opens a drawing from the recent files list (File > Recent Files).

**OPTIONS** `CURSOR, DDSELECT, OP` Cursor and selection settings: crosshair size and color, pickbox, snap marker, grips, rollover tips.

**ORTHO** Turns ortho mode on or off.

**OSNAP** `DDOSNAP, DS, DSETTINGS, OS` Sets running object snap modes (settings panel).

**-OSNAP** Sets running object snap modes at the command line.

**OVERKILL** Removes duplicate and overlapping objects.

## P

**PAGESETUP** `PAGESETUPS` Sets the current layout's paper size, orientation and units.

**PAN** `P` Shifts the view.

**PASTEBLOCK** Pastes the clipboard as one block (Ctrl+Shift+V).

**PASTECLIP** Pastes the clipboard at a point you pick (Ctrl+V).

**PASTEORIG** Pastes the clipboard at the same coordinates it was copied from.

**PDFATTACH** Attaches a page of a PDF file to the drawing.

**PEDIT** `PE` Edits polylines.

**PLAN** Back to the plan (Top) view.

**PLINE** `PL` Creates a 2D polyline of line and arc segments.

**PLOT** `PLT, PRINT` Plots the drawing to a PDF file or a printer.

**-PLOT** Plots at the command line (for scripts).

**POINT** `PO` Creates a point object.

**POLYGON** `POL` Creates a regular polygon.

**POLYHEDRON** `PLATONIC` Creates a Platonic solid (tetrahedron, cube, octahedron, dodecahedron, icosahedron) or a truncated icosahedron (soccer ball).

**PRESSPULL** `PRESS` Click inside a closed area or on a flat face of a solid (or pick an Edge) and pull it into a solid.

**PROPERTIES** `CH, DDCHPROP, DDMODIFY, MO, PR, PROPS` Shows the Properties palette.

**PSPACE** `PS` Goes back to the layout's sheet (paper space) from a viewport.

**PURGE** `PU` Removes unused layers and block definitions.

## Q

**QSAVE** Saves the current drawing.

**QUICKCALC** `QC` Opens QuickCalc: a calculator, unit converter, and geometry from the drawing.

**QUICKPROPERTIES** `QP, QUICKPROP` Quick Properties: a small panel with the main properties of the selected objects.

**QUIT** `EXIT` Exits the program (offering to save each changed drawing).

## R

**RECTANG** REC, RECTANGLE Creates a rectangular polyline.

**REDO** MREDO Reverses the effects of the previous UNDO.

**REGEN** R, RE, REDRAW, REDRAWALL, REGENALL Regenerates the display.

**REINIT** Reloads the command alias (.pgp) file.

**RENAME** DDRENAME, REN Renames a block, layer, text style or dimension style.

**REVCLOUD** RC Creates a revision cloud.

**REVOLVE** REV Revolves shapes about an axis to make solids (or surfaces from open shapes).

**REVSURF** Creates a revolved surface mesh: a curve spun about an axis (SURFTAB1 x SURFTAB2 faces).

**ROTATE** RO Rotates objects around a base point.

## S

**SAVE** Saves the drawing.

**SAVEAS** Saves the drawing under a new name.

**SAVEASR12** Saves an AutoCAD R12 DXF (no lineweights or hatches).

**SCALE** SC Enlarges or reduces objects.

**SCRIPT** SCR Runs a sequence of commands from a script (.scr) file.

**SELECTALL** AI_SELALL Selects every visible object.

**SELECTSIMILAR** SSIM Selects all objects of the same type and layer as the selection.

**SENDTOBACK** Draws the selected objects under everything else.

**SETVAR** SET Lists or changes the values of system variables.

**SLICE** SL Cuts solids with a plane, keeping one side or both.

**SNAP** SN Sets snap spacing or turns snap on/off.

**SOCCERBALL** TRUNCICOSA Creates a truncated icosahedron (a soccer ball): 12 pentagons and 20 hexagons.

**SPELL** SP Checks the spelling of text in the drawing.

**SPHERE** Creates a 3D solid sphere.

**SPLINE** SPL Draws a smooth curve through (Fit) or controlled by (CV) the points you pick.

**SSX** Express: selects objects matching a picked object's type, layer and color.

**STLIN** IMPORT3D, OBJIN Brings an STL or OBJ file (from a 3D model or a printing site) in as a solid.

**STLOUT** Saves solids as an STL file for 3D printing.

**STRAYS** FINDSTRAYS Finds objects lying far from the rest of the drawing, or far too big for it, to move back or erase.

**STRETCH** S Stretches objects crossed by a selection window.

**STRUTS** STRUT, WIRESOLID Turns the lines of a wireframe into solid bars the length of each line, with a rectangle, polygon, circle or drawn-polyline section. Polylines with arcs, arcs, circles and splines get one strut that follows the curve.

**STYLE** DDSTYLE, ST Creates, modifies, or sets the current text style.

**-STYLE** Text styles at the command line.

**SUBTRACT** SU Removes solids from other solids.

**SURFTAB1** Number of faces around a revolved surface (REVSURF).

**SURFTAB2** Number of faces along a curve of a revolved surface (REVSURF).

**SWEEP** Sweeps shapes along a path (line, arc, polyline, spline, helix) to make solids or surfaces.

**SYSWINDOWS** Lists the open drawings.

## T

**TCASE** CHANGECASE Changes the case of text, MText and dimension text: Sentence, lower, UPPER, Title, tOGGLE.

**TCOUNT** Express: numbers selected text objects.

**TETRAHEDRON** TETRA Creates a tetrahedron: 4 triangular faces.

**TEXT** DT, DTEXT Creates single-line text, typed in place on the drawing.

**TEXTEDIT** DDEDIT, ED, MTEDIT Edits the contents of a text object (in place on the drawing).

**TEXTTOFRONT** Brings text, dimensions and leaders in front of everything else.

**THICKNESS** TH Sets the thickness (extrusion height) for new lines, arcs, circles and polylines.

**TOOLBAR** TB, TO Shows the list of toolbars, to turn them on or off (drag a toolbar by its dotted grip to dock it on any edge or float it).

**TOOLBAROPTIONS** TBOPTIONS Opens the Toolbar Options window: icon size, line weight, spacing and colors.

**TORUS** TOR Creates a 3D solid torus (a ring), lying in the working plane.

**TRIM** TR Trims objects to meet the edges of other objects.

**TXT2MTXT** T2M Express: combines single-line text into one MTEXT object.

**TXTEXP** EXPLODETEXT, TEXTEXPLODE Explodes text into closed polylines tracing each letter's outline (Express Tools).

## U

**U** Reverses the most recent operation.

**UCS** WORKPLANE, WP Sets the working plane: through three points, on an object's plane, at an origin, turned about an axis, facing the view, or back to World.

**UNDO** Reverses the effect of commands.

**UNION** UNI Joins solids into one.

**UNITS** DDUNITS, UN Sets the drawing's length and angle units, precision and insertion units.

## V

**-VIEW** V, VIEW, VIEWS Changes to a standard view: Top, Bottom, Left, Right, Front, Back or an isometric view.

**VPORTS** VIEWPORTS, VPORT Splits the Model tab into several viewports, each with its own view: Single, Join, 2, 3, 4.

**VPSCALE** VPS Sets the scale of layout viewports (1:2, 1/4"=1'-0", 2:1, Fit ...).

**VSCURRENT** SHADEMODE, VISUALSTYLES, VS Sets the visual style: Wireframe, Hidden, Shaded or Shaded with Edges.

## W

**WEDGE** WE Creates a 3D solid wedge: full height at the first corner, sloping to the opposite side.

## X

**XLIST** Express: lists the properties of an object nested in a block.

## Z

**ZOOM** Z Changes the view magnification.

CHAPTER 21

# When Something Goes Wrong

*Common problems and what to do.*

| | |
|---|---|
| **Fancy Pencil doesn't start** | Open a terminal and type `fancypencil` to see the message. When started from the menu, messages go to `~/.cache/fancypencil.log`. |
| **It seems stuck** | Press `Esc` a few times. If it still doesn't answer, type `pkill -f fancypencil` in a terminal; the next start offers your safety copies. |
| **A .dwg file won't open** | Install ODA File Converter (Chapter 2): Help > Get ODA File Converter. |
| **The .deb opens in Archive Manager** | Right-click it > Open With... > App Center. |
| **Clicks land in the wrong place** | An object snap may be taking the point: press `F3` to turn OSNAP off, or type `NON` before the point. |
| **Lines won't go at an angle** | ORTHO is on: press `F8`. |
| **I can't select or change an object** | Its layer may be locked or frozen: check the Layer Properties Manager. |
| **The dashes of a linetype don't show** | Change LTSCALE: larger for a big drawing, smaller for a small one. |
| **Text looks wrong in AutoCAD** | Use fonts AutoCAD has too, or send AutoCAD the font. |
| **A command did the wrong thing** | `U` or `Ctrl+Z` undoes it; one U undoes a whole command. |
| **I lost my work in a crash** | File > Recover Safety Copies. Saving also keeps the previous version as name.bak. |
| **Out of memory on a big operation** | Fancy Pencil stops that command and keeps running. Work on fewer objects at once; U undoes any part done. |

## Your files and settings

| | |
|---|---|
| `~/.config/fancypencil/` | Settings, recent files, aliases, toolbars, plot settings, views. |
| `~/.local/share/fancypencil/autosave/` | Safety copies. |
| `~/.local/share/fancypencil/logs/` | Command history logs (when turned on). |
| `~/.cache/fancypencil.log` | Start-up and error messages. |
| `/opt/fancypencil/` | The program itself (installed by the .deb). |